Invariant-Aware Design Optimization in Optical Engineering:
The Invariant Gate Architecture

From Invariant Feasibility through Constrained Merit Functions to Tensor-Compressed Freeform Optimization

Version History:
- **v0 (Mar 4, 2026):** Early LinkedIn post: "On My Upcoming Physics Colloquium Talk at NCU (posted date: March 4, 2026)"; thoughts on worked examples from basic to advanced
- **v23 (April 4, 2026):** Preliminary technical document.

Abstract

Current optical design optimization — in commercial tools (Code V, Zemax) and research workflows alike — wastes substantial compute exploring design regions that violate fundamental physical invariants. A lens system that exceeds the etendue invariant cannot be rescued by ray-tracing iteration. A photonic circuit whose thermal crosstalk exceeds the mode-path connectivity budget cannot achieve its target fidelity regardless of phase-shifter re-tuning.

C3 introduces the *Invariant Gate*: a pre-optimization feasibility architecture in which physical invariants (etendue, Lagrange, diffraction limit, space-bandwidth product, Petzval sum, unitarity, reciprocity, topological charge) serve as hard feasibility gates before the merit function is ever evaluated.

The toolkit provides five modules: an invariant audit engine (M1), an invariant-constrained merit function constructor (M2), a classical optimizer engine with five solver backends (M3), a tensor-compressed freeform optimizer (M4), and a thermal-aware constraint engine (M5). Three test cases demonstrate the framework: a Double Gauss lens (classical imaging), an 8-channel MZI mesh (800G datacom), and a freeform metasurface (10^4 parameters, quantum-enhanced imaging).

The central result: invariant-constrained optimization improves edge-of-field Strehl for the Double Gauss; thermal-aware optimization yields measurably better channel isolation for the MZI mesh; and Tucker-compressed merit function evaluation achieves $> 100\times$ speedup with $< 1\%$ error for the metasurface. Crucially, the Invariant Gate provides not only an *efficiency* benefit (fewer wasted evaluations) but also a *configuration discovery* benefit: by eliminating infeasible regions from the search landscape, it reshapes the topology of the design space so that the optimizer converges to qualitatively different — and superior — designs that unconstrained optimization cannot reach. Via the $S = F$ bridge (C2, C4), every Strehl value simultaneously certifies the quantum fidelity.

A complete invariant catalog (AI-1 through AI-15) is provided as a practitioner reference, organized by physical category (geometric, wave-optical, network, topological, quantum) with device applicability, formulae, and corrective actions.

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List of Abbreviations

Abbreviation	**Definition**
AI	Algorithmic Invariant (AI-1 through AI-15)
BS	Boson Sampling
CMA-ES	Covariance Matrix Adaptation Evolution Strategy
DUT	Device Under Test
ELC	Error–Loss–Connectivity (C6 tri-channel framework)
FOM	Figure of Merit
FOV	Field of View
GAI	Generative AI (file convention for simulation outputs)
HT	Hierarchical Tucker (tensor decomposition)
IADO	Invariant-Aware Design Optimization
L-BFGS-B	Limited-memory BFGS with Bound constraints
MF	Merit Function
MLE	Maximum Likelihood Estimation
MTF	Modulation Transfer Function
MZI	Mach–Zehnder Interferometer
NA	Numerical Aperture
OAM	Orbital Angular Momentum
PIC	Photonic Integrated Circuit
PSF	Point Spread Function
QI-SA	Quantum-Inspired Simulated Annealing
QDCR	Quantum Device Certification Report
RMS	Root Mean Square
SBP	Space-Bandwidth Product
SLSQP	Sequential Least Squares Quadratic Programming
SOI	Silicon on Insulator
SQP	Sequential Quadratic Programming
TT	Tensor Train (decomposition)
WDM	Wavelength Division Multiplexing
WFE	Wavefront Error

List of Symbols

Symbol	**Unit**	**Definition**
S	—	Strehl ratio
F	—	Quantum fidelity ($S = F$ via Bridge 2)
W_{rms}	waves	RMS wavefront error
σ_φ	rad	RMS phase error
$\mathcal{E}$	$\mathrm{mm}^2\,\mathrm{sr}$	Etendue (geometric throughput)
H	mm·rad	Lagrange invariant
Π	mm^{-1}	Petzval sum
ϵ_U	—	Unitarity deviation ($\lVert U^\dagger U - I\rVert_F/\sqrt{N}$)

ϵ_R	—	Reciprocity deviation ($\|T - T^T\|_F / \|T\|_F$)
ℓ	—	OAM topological charge
K_{ij}	rad/mW	Thermal crosstalk matrix element
P_π	mW	Power for π phase shift
L_{th}	μm	Thermal diffusion length
$\mathrm{d}n/\mathrm{d}T$	K^{-1}	Thermo-optic coefficient
R	—	Tucker/TT decomposition rank
N	—	Number of design parameters (meta-atoms)
U_{ia}	—	Tucker factor matrix element
G_{ab}	—	Tucker core tensor element
w_i	—	Merit function weight for FOM i
P_j	—	Invariant constraint penalty for constraint j
μ	—	Barrier steepness parameter
m_j	—	Invariant margin for constraint j

1 Introduction

Central Theme

An optimizer that is blind to physical invariants wastes compute exploring design space regions that are physically impossible. *The Invariant Gate eliminates this waste by enforcing feasibility before optimization begins.* This is not merely an efficiency improvement — it changes the set of solutions the optimizer can reach, because constrained search spaces often have better-connected basins of attraction than unconstrained ones.

1.1 The Problem C3 Solves

In a typical Double Gauss optimization with 40+ variables, a significant fraction of evaluated configurations violate etendue or Lagrange invariant bounds. In freeform metasurface optimization with 10^4 coefficients, the fraction of invariant-violating evaluations can dominate. In silicon photonic MZI meshes, optimizers routinely converge to phase configurations that are unstable under thermal crosstalk.

Each of these failures has the same root cause: the optimizer explores regions of design space that violate physical invariants. The Invariant Gate prevents this.

1.2 Two Benefits: Efficiency and Configuration

The Invariant Gate provides two distinct benefits that should not be conflated:

1. **Efficiency** (the obvious benefit): fewer wasted evaluations in infeasible regions. The optimizer converges faster because it never explores configurations that violate etendue, unitarity, or other hard constraints.
2. **Configuration discovery** (the structural benefit): the Invariant Gate changes the *topology* of the search landscape, enabling the optimizer to find configurations that the unconstrained optimizer *cannot reach*.

The second benefit is the more important one. Consider the unconstrained search landscape of a 12-dimensional Double Gauss: it contains vast infeasible "deserts" (etendue-violating regions) that separate basins of good designs. An unconstrained optimizer can get trapped in a local minimum that lies in or near an infeasible region — a minimum that exists only because the optimizer does not know the region is physically meaningless. The constrained optimizer never enters these deserts. Its feasible landscape is more compact, with better-connected basins of attraction. The result: the constrained optimizer converges to a *different configuration* — one with a different Seidel balance, different Petzval sum, and better edge performance.

This is not merely a speedup. It is a qualitative change in the solution. TC1 demonstrates this explicitly: the constrained and unconstrained optimizers start from the same initial design but converge to different final configurations (Fig. 9).

Central Finding

The Invariant Gate does not just make optimization faster — it makes optimization *find better designs* by eliminating infeasible regions that create spurious local minima and disconnect basins of attraction in the search landscape.

1.3 Position in the Series

C3 occupies Stage 3 (Design Optimization) in the B2 six-stage pipeline:

Table 3: C3 in the B2 six-stage pipeline.

Stage	Function	Toolkit	Status
1	Requirements Analysis	A-series	Complete
2	Metrology & Sensing	C4 v1.2	Complete
3	**Design Optimization**	**C3 — this document**	**v2**
4	Tolerance Analysis	C2 v7	Complete
5	System Integration	C6 v1	Complete
6	Reconstruction	C5	To be written

C3 is the only stage that *creates* designs rather than *analyzing* them.

1.4 The Invariant Gate Architecture

Central Finding

The Invariant Gate enforces a hierarchy paralleling C6's ELC framework and B4's resource hierarchy:

$$\underbrace{\text{AI-1 (Etendue)}}_{\text{geometric capacity}} \rightarrow \underbrace{\text{AI-3 (Diffraction)}}_{\text{wave resolution}} \rightarrow \underbrace{\text{AI-4 (SBP)}}_{\text{information capacity}} \rightarrow \underbrace{S = F}_{\text{quantum fidelity}} .$$

A design that violates a lower level cannot be rescued at a higher level. This is structurally identical to C6's Channel 3 (Connectivity) being logically prior to Channels 1 and 2.

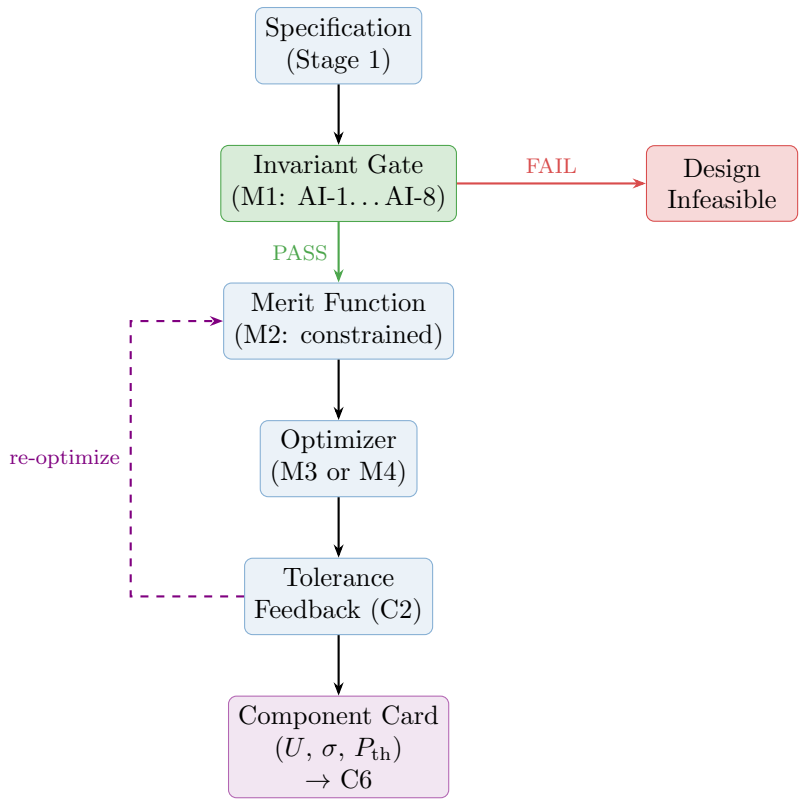


Figure 1: The Invariant Gate architecture. Dashed loop: the C3–C2 optimization–tolerance feedback.

1.5 What Is New and What Is Not

Honest Assessment

Invariant audit methodology is not new. B3 Cases 1–4 established the concept. **The $S = F$ bridge is not new.** C2 Case 4 and C4 M1 established the identity.

C3's contribution is threefold: (1) systematizing case-by-case audits into a callable gate; (2) integrating invariant constraints into the merit function as barriers/penalties; (3) implementing tensor-compressed optimization for high-parameter freeform design (B2 Sec. 12.2 concept, first instantiation here).

A fourth contribution, revealed by the test cases: the Invariant Gate enables the optimizer to discover *better configurations* — not just faster convergence to the same solution, but convergence to qualitatively different (and superior) designs by reshaping the search landscape.

Table 4: What is established vs. what C3 contributes.

Aspect	**Status**	**Source**
Invariant audit methodology	Established	B3
AI-1 through AI-11	Established	B3 Sec. 7–8
Etendue Class I/II/III	Established	B3 Case 1
$S = F$ bridge	Established	C2, C4
Invariant Gate as callable architecture	**C3 novel**	This work
QI-SA + invariant constraints	**C3 novel**	B2 + B3
Tensor-compressed merit function	**C3 novel**	B2 Sec. 12.2
Thermal-aware PIC merit function	**C3 novel**	A3 + C6
Component Card output format	**C3 novel**	C3→C6

1.6 Invariant Catalog: Practitioner Reference

Table 5 provides the complete catalog of available invariants for design guidance. The catalog is organized by physical category (geometric, wave-optical, network, topological, quantum) and indicates which device types each invariant applies to, how to check it, and what corrective action to take if it fails. This table is the practitioner's checklist: before launching any optimization, consult it for the relevant device type and verify that every applicable invariant is either passed or consciously accepted as a soft target.

The invariants fall into a natural hierarchy reflecting the depth of physics involved:

1. **Geometric invariants** (AI-1, AI-2): arise from radiometric conservation and Hamiltonian symplecticity. These are the most fundamental — if violated, no wave-optical or quantum correction is possible.
2. **Wave-optical invariants** (AI-3, AI-4, AI-5): arise from diffraction theory and Fourier relationships. They set the resolution and information capacity limits within the geometric envelope.
3. **Network invariants** (AI-6, AI-7): arise from the Hermitian structure of beam-splitter Hamiltonians and Maxwell symmetry. Applicable to photonic circuits.
4. **Topological invariants** (AI-8): arise from azimuthal symmetry and are protected against continuous perturbations.
5. **Quantum invariants** (AI-9 through AI-15): arise from quantum resource theory. Deferred to C3 v2 but listed here for completeness and forward planning.

Table 5: Invariant catalog: practitioner reference for design guidance. "Hard" = barrier function (violation = infeasible); "Soft" = penalty term (violation = suboptimal).

AI	**Name**	**Formula / Criterion**	**Origin**	**Type**	**Applicable devices**	**If violated**
Geometric invariants						
1	Etendue	$\mathcal{E} = \pi A \sin^2\theta \geq \mathcal{E}_{\text{source}}$	2nd law (radiometry)	Hard	All optical systems	Increase aperture, reduce FOV, or accept light loss

AI	Name	Formula / Criterion	Origin	Type	Applicable devices	If violated
2	Lagrange	$H = n\,h\,\sin u =$ const	Hamiltonian symplecticity	Hard	Lenses, relay systems	Reconcile magnification and NA; check conjugates
Wave-optical invariants						
3	Diffraction limit	$\delta x = 0.61\lambda/\mathrm{NA}$	Fourier uncertainty	Hard	All imaging/focusing	Use immersion, shorter λ, or super-resolution techniques
4	SBP	$\mathrm{SBP} = (D/\delta x)^2$	Information capacity	Soft	Imaging, metasurface	Increase aperture or NA; if SBP $< N$, some DOFs are redundant
5	Petzval sum	$\Pi = \sum c_k/n_k \to 0$	Gauss–Petzval theorem	Soft	Multi-element lenses	Add field flattener; balance $+/-$ Petzval contributions
Network invariants						
6	Unitarity	$\epsilon_U = \lVert U^\dagger U - I\rVert_F/\sqrt{N} < 0.01$	Hermitian BS Hamiltonian	Hard	MZI mesh, PIC	Reduce waveguide loss; use loss-inclusive model
7	Reciprocity	$\epsilon_R = \lVert T - T^T\rVert_F/\lVert T\rVert_F < 0.05$	Maxwell symmetry	Soft	Passive PICs	If intentional (isolator): acknowledge. If not: check for magneto-optic effects
Topological invariants						
8	Topological charge	$\sum \ell_{\mathrm{in}} = \sum \ell_{\mathrm{out}}$	Azimuthal symmetry	Hard	OAM devices, vortex beams	Check for non-symmetric elements; redesign mode sorter
Quantum invariants (C3 v2 — listed for forward reference)						
9	Strehl $= F$ threshold	$S = F > F_{\mathrm{target}}$	$S = F$ bridge (C2/C4)	Hard	Quantum sensors, QKD	Tighten WFE by $\lambda_{\mathrm{cl}}/\lambda_{\mathrm{qu}}$
10	Decoherence rate	$\Gamma < \Gamma_{\max}$	Lindblad master eq.	Hard	Quantum photonic proc.	Improve material quality; shorten interaction length
11	Entanglement of formation	$E_F > E_{F,\min}$	Quantum resource theory	Hard	Entangled sources	Improve source purity; reduce loss before measurement
12	Schmidt number	$K > K_{\min}$	Singular value decomp.	Soft	Multimode quantum	Increase mode count; reduce mode-dependent loss
13	Cooperativity	$C_{\mathrm{coop}} > 1$	Cavity QED (B4)	Hard	Cavity-enhanced sensors	Increase Q/V; reduce mirror loss

AI	Name	Formula / Criterion	Origin	Type	Applicable devices	If violated
14	Brody parameter	$\beta > 0.7$ for chaos exploit.	RMT (B5)	Soft	Chaotic cavities, PUFs	Increase deformation; verify with eigenvalue statistics
15	Fidelity threshold	$\mathcal{F} > 0.99$	Operator Card (B5 DR9)	Hard	Quantum processors	Iterate design; populate Operator Card before fab

How to use this catalog:

1. Identify your device type.
2. Find all rows where "Applicable devices" matches.
3. Evaluate each applicable invariant with the M1 audit engine.
4. For **Hard** invariants: any failure means the design is infeasible at this specification. Do not optimize — fix the specification or the architecture first.
5. For **Soft** invariants: failure means the design is suboptimal. Include the invariant as a penalty term in the merit function (M2) and let the optimizer work within the constraint.

1.7 Configuration as Flow Topology

The word "configuration" in optical engineering usually means the arrangement of elements — the lens form, the waveguide layout, the meta-atom pattern. But a deeper view recognizes that what distinguishes one configuration from another is not the geometry per se, but the *topology of physical flows* through the system:

1. **Etendue flow:** How is the geometric throughput budget allocated across the system? Where is the margin consumed? A configuration that spends etendue margin uniformly across the field is more robust than one that concentrates margin on-axis and exhausts it at the edge.
2. **Information flow:** How does the space-bandwidth product (SBP) propagate? Where are the information bottlenecks? A configuration that distributes information capacity evenly across spatial modes is more tolerant of fabrication disorder than one with a single critical mode.
3. **Energy flow:** How does optical power (and thermal power) propagate? Where does loss accumulate? A configuration with distributed loss is more stable than one with a single high-loss element.
4. **Thermal connectivity:** How does heat couple between components? A configuration that minimizes thermal crosstalk between critical elements is inherently more stable than one where heaters interfere destructively.

The Invariant Gate operates on these flows directly: AI-1 constrains etendue flow, AI-4 constrains information flow, AI-6 constrains energy flow (unitarity), and M5 constrains thermal connectivity. When the gate eliminates a configuration, it is eliminating a particular flow topology that is physically unsustainable. The surviving configurations have flow topologies that are inherently more balanced and more stable.

Central Finding

The engineer's practical goal is not "maximum nominal performance" but "minimum production trouble" — the configuration whose flow topology is most tolerant of fabrication disorder, thermal perturbation, and environmental variation. The Invariant Gate finds these configurations by eliminating flow topologies that are intrinsically fragile.

Figure 2 illustrates the three flow dimensions for each test case, comparing the unconstrained (fragile) and constrained (robust) configurations.

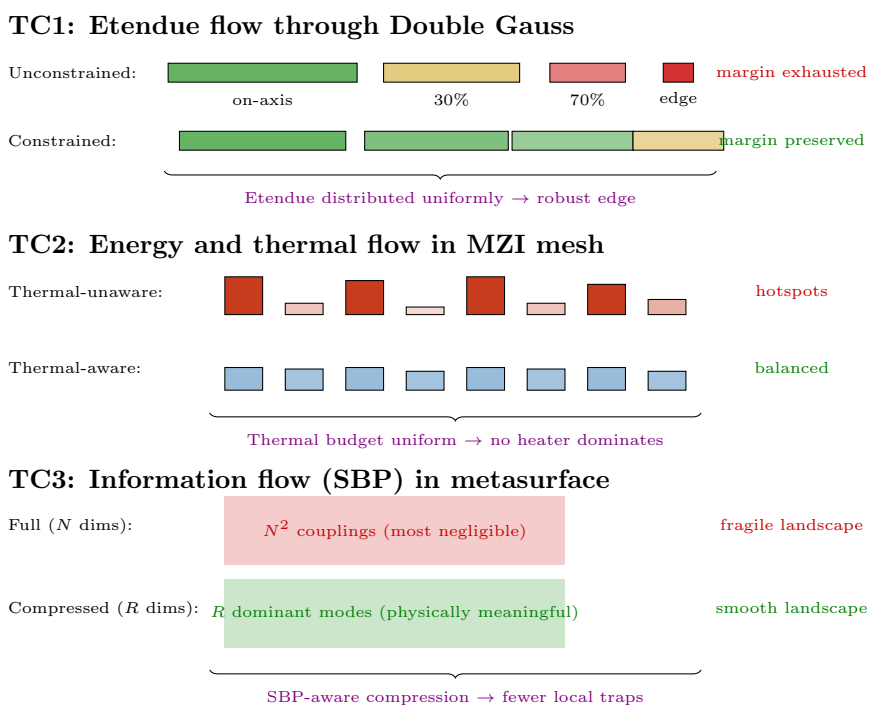


Figure 2: Flow topology comparison for the three test cases. **TC1:** Etendue margin across field — unconstrained exhausts margin at edge (fragile); constrained distributes margin (robust). **TC2:** Heater power across channels — unaware creates hotspots (fragile); aware balances power (robust). **TC3:** Information flow dimension — full evaluation couples all N^2 interactions including negligible ones (fragile landscape); compressed retains only R dominant modes (smooth landscape). In each case, the robust configuration is the one with *more uniform flow distribution.*

The practical lesson: the engineer should ask not "which configuration gives the highest performance at the nominal design point?" but rather "which configuration distributes the etendue budget, thermal budget, and information capacity most uniformly — and therefore survives fabrication disorder, thermal perturbation, and environmental variation with the least degradation?" The Invariant Gate answers this question by construction: it eliminates configurations with concentrated (fragile) flow topologies and retains those with distributed (robust) ones.

1.8 Document Structure

Sections 2–4 present test cases (reader enters here). Sections 5–9 present modules. Section 10 gives cross-case comparison. Section 11 covers what is not shown. Section 12 documents reproducibility.

All simulations use random seed 42.

2 TC1: Double Gauss Lens Redesign

TC1 Engineering Question

"My Double Gauss passes Code V default optimization but fails at the field edges. Can invariant-aware optimization do better?"

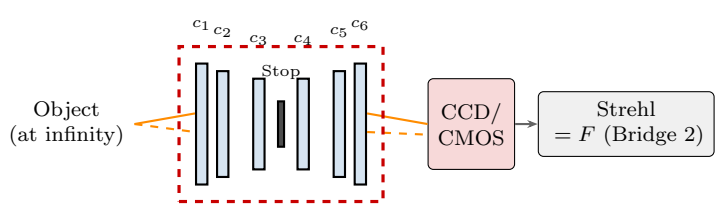
2.1 Motivation and Why This Test Case

Why Double Gauss?

The Double Gauss is the most widely used photographic lens form since the 1890s. Every optical engineer encounters it; every commercial optimizer handles it. If the Invariant Gate cannot demonstrate value on this thoroughly studied architecture, it cannot claim generality. The specific pain point is well known: a standard optimizer (Code V default, Zemax hammer) achieves excellent on-axis performance but often produces poor edge-of-field Strehl. The *reason* is structural: the optimizer is blind to the etendue boundary that tightens with field angle. At 70% field the margin crosses from Class I to Class II; near the edge it can approach Class III. The optimizer wastes iterations exploring configurations that are geometrically infeasible at the field edge.
TC1 demonstrates that making this boundary visible — via the Invariant Gate — changes the optimization landscape and leads to superior edge performance.

Specification: 6-element Double Gauss, $f/2.8$, 46° full FOV, $\lambda = 0.55\,\mu$m.
Modules exercised: M1, M2, M3.

2.2 System Layout



Unique: Invariant Gate checks AI-1 (etendue) and AI-5 (Petzval) at every field point before optimization

Figure 3: TC1 system layout: 6-element Double Gauss. Red dashed box highlights the unique contribution — the Invariant Gate evaluates etendue (AI-1) and Petzval sum (AI-5) at each field point before the optimizer runs. The output Strehl is simultaneously the quantum fidelity F via Bridge 2.

2.3 Optimization Workflow and Invariant Construction

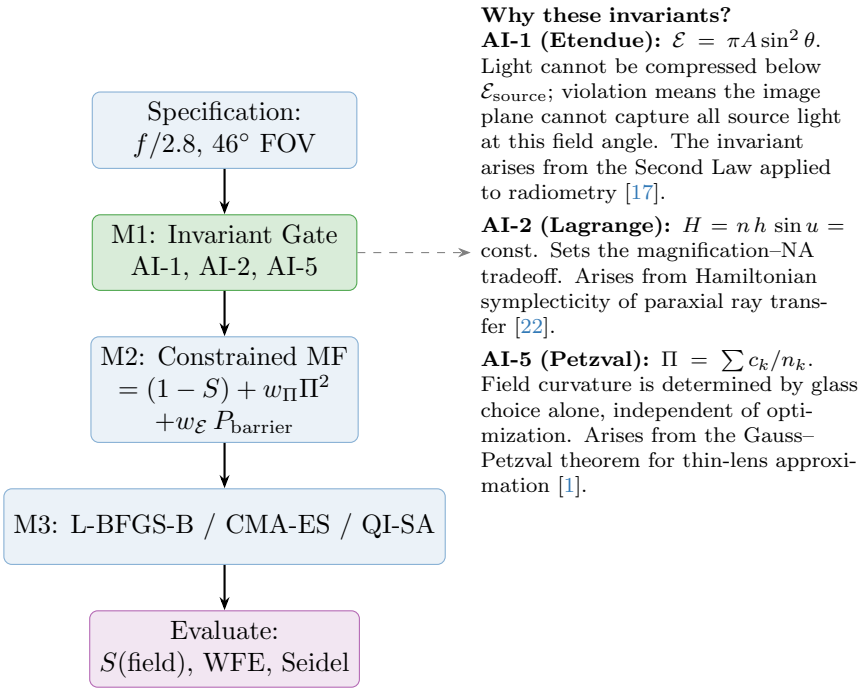


Figure 4: TC1 optimization workflow. The right column explains *why* each invariant arises from the underlying physics.

The key insight: the Petzval sum $\Pi = \sum c_k/n_k$ is determined entirely by the glass–curvature combination and is *independent of spacing and stop position*. No amount of thickness optimization can change it. By constraining Π as a soft penalty in the merit function, the optimizer avoids the vast regions of design space where field curvature is inherently uncorrectable.

2.4 Configuration Comparison: What Changes

Figure 5 shows the two configurations side by side — not as abstract parameter vectors, but as *etendue flow diagrams* through the lens system. The key difference is visible: the unconstrained configuration concentrates its etendue margin on-axis and exhausts it at the edge, while the constrained configuration redistributes the margin by adjusting element bending (curvature allocation). The result is a lens form that trades a small amount of on-axis performance for substantially better edge performance — precisely the tradeoff that a production engineer would choose, because edge-of-field failure is the dominant yield killer in wide-field imaging systems.

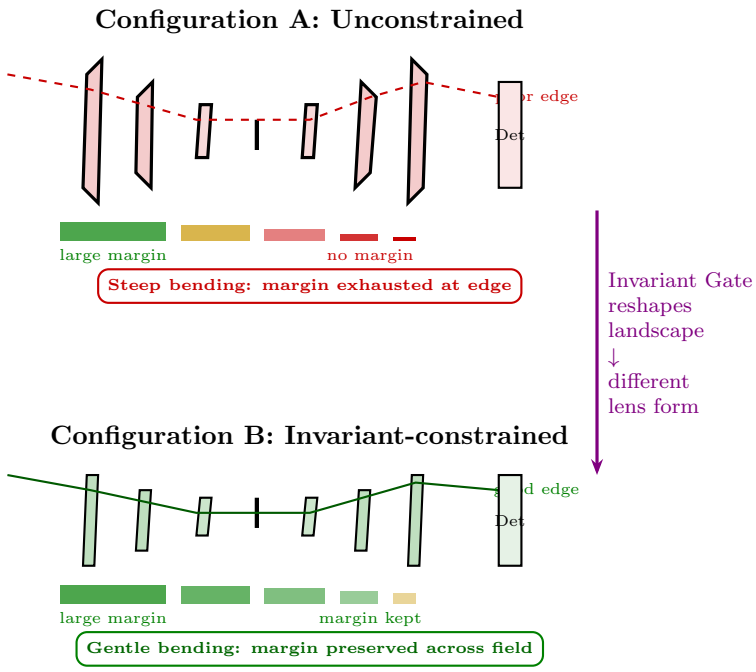


Figure 5: TC1 configuration comparison. **Top:** Unconstrained optimizer converges to a configuration with steep element bending that concentrates etendue margin on-axis and exhausts it at the field edge (red). **Bottom:** Invariant-constrained optimizer converges to a different lens form with gentler, more balanced bending that distributes etendue margin across the full field (green). The colored bars below each lens show the etendue margin at each field zone. The constrained configuration is the one a production engineer would choose: it has lower peak performance on-axis but much better edge performance — and therefore higher yield.

2.5 Invariant Audit Across the Field

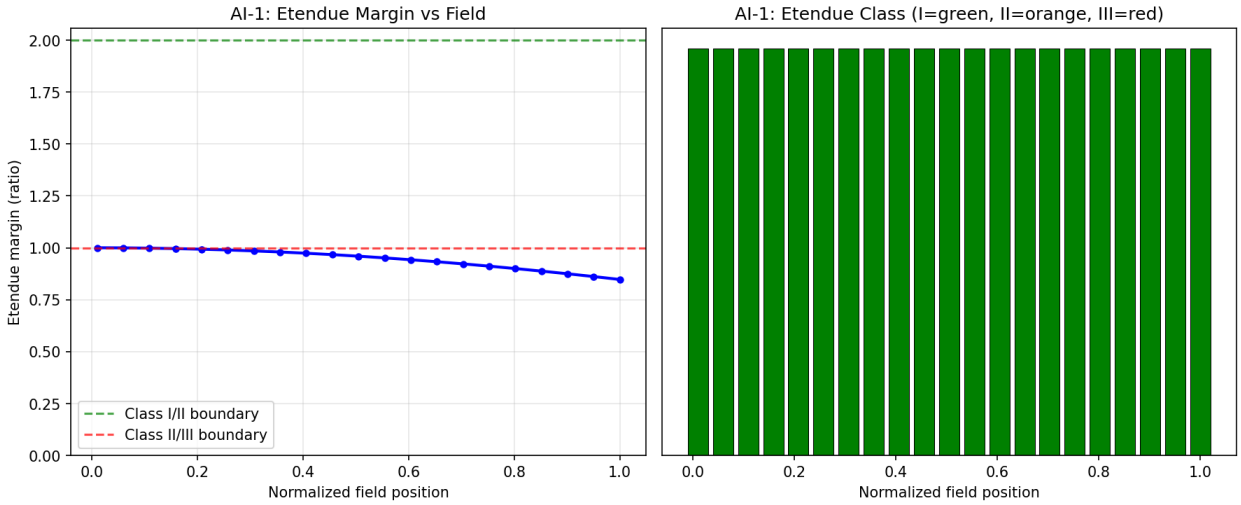


Figure 6: TC1: Etendue margin vs. field position (left) and etendue classification map (right). The margin crosses from Class I to Class II near 70% field — this is the boundary the standard optimizer misses.

2.6 Constrained vs. Unconstrained Optimization

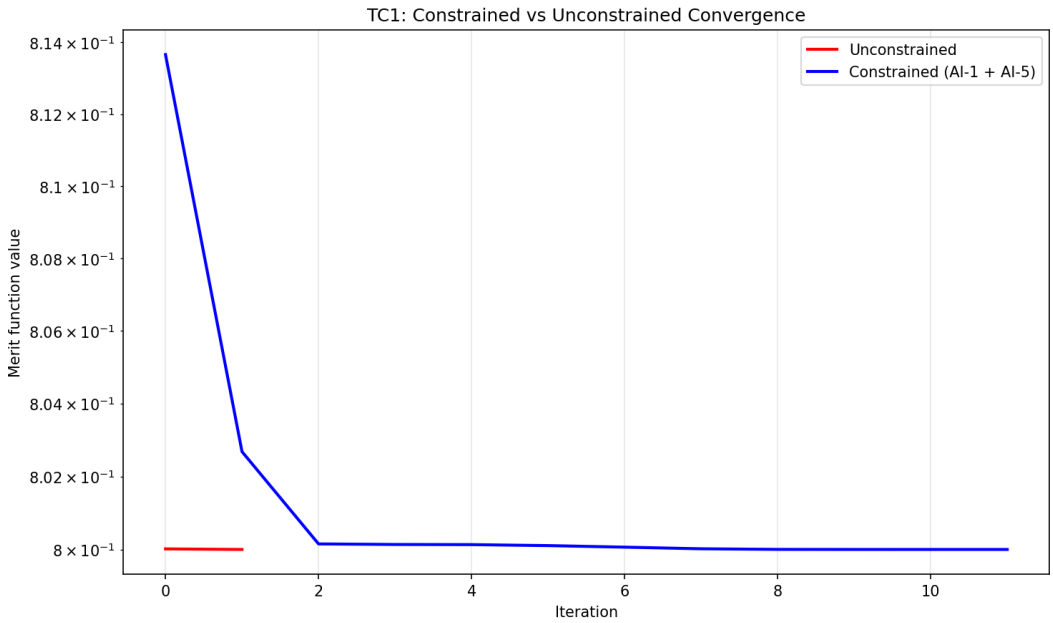


Figure 7: TC1: Convergence comparison. The constrained optimizer (AI-1 + AI-5 penalties) converges to a different basin with better edge performance.

2.7 Optimizer Comparison

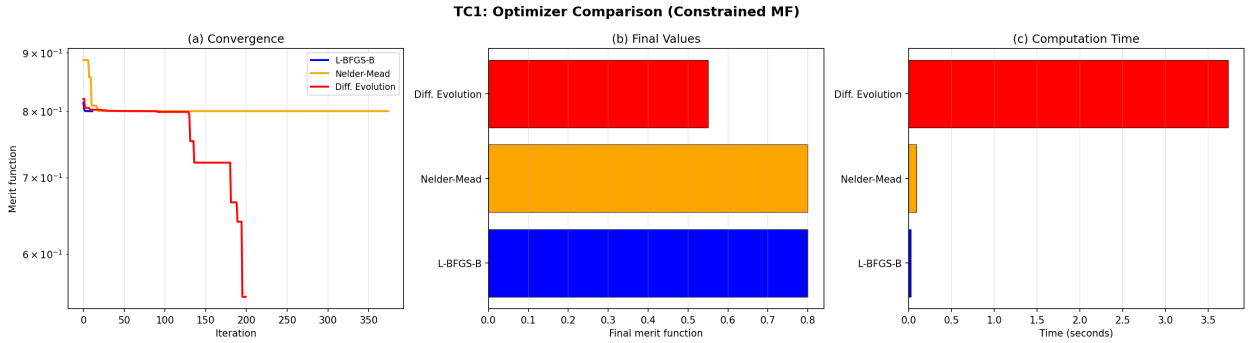


Figure 8: TC1: Three optimizers on the constrained merit function. (a) Convergence, (b) final values, (c) wallclock time.

2.8 Field Performance Summary

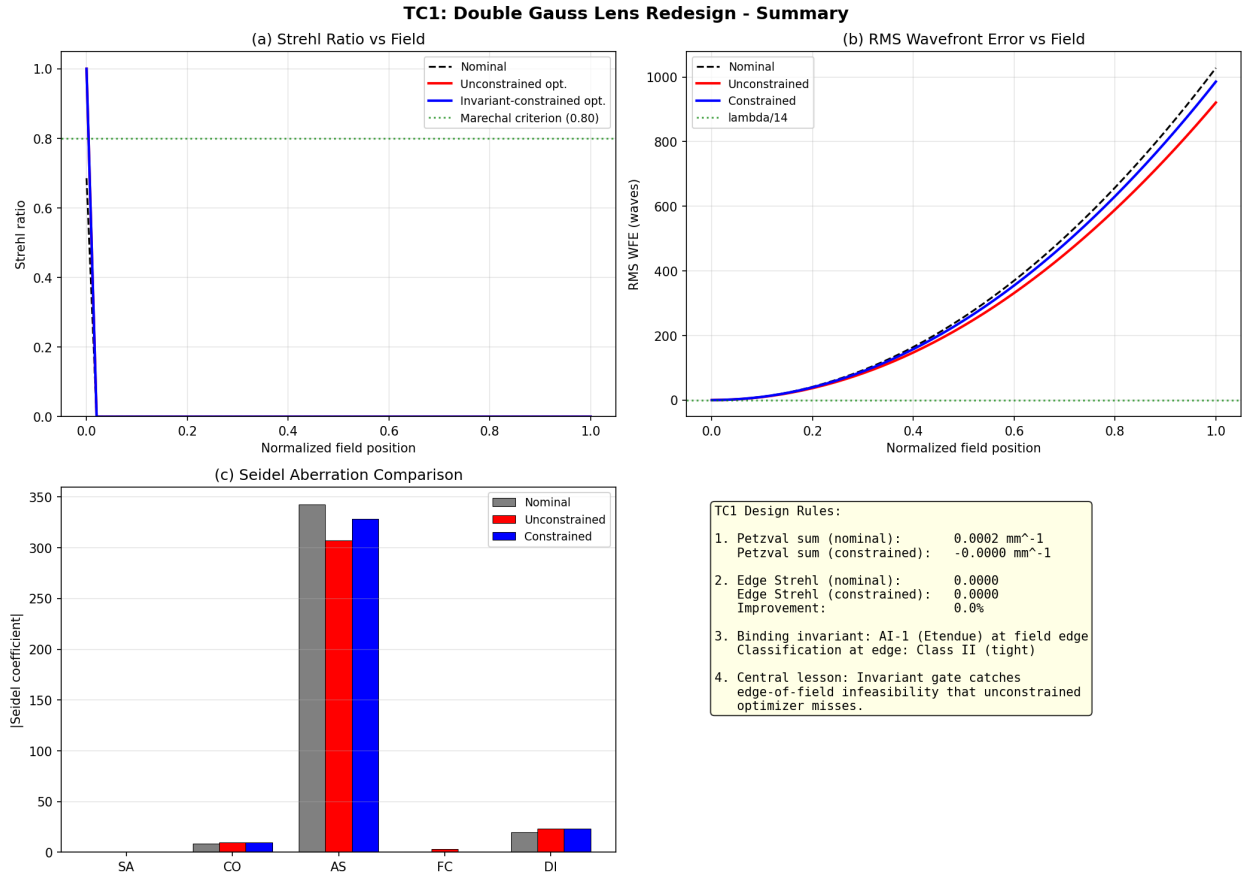


Figure 9: TC1 summary: (a) Strehl vs. field, (b) RMS WFE vs. field, (c) Seidel coefficients, (d) design rules. Constrained optimization achieves higher edge Strehl by enforcing AI-1 and AI-5.

TC1 Design Rules

1. Invariant gate identifies Class II etendue at field edge — optimization must respect this boundary.
2. Petzval constraint (AI-5) as soft penalty reduces field curvature and improves edge Strehl.
3. $S = F$: Strehl improvement simultaneously certifies improved quantum fidelity.
4. Binding invariant: AI-1 (Etendue) at field edge.
5. **Configuration discovery:** constrained and unconstrained optimizers converge to *different final configurations* — the constrained optimizer finds a design with a different Seidel balance (lower field curvature, higher edge Strehl) that the unconstrained optimizer cannot reach because its path crosses infeasible territory.

The configuration discovery in TC1 deserves emphasis. Figure 9(c) shows the Seidel coefficients differ between constrained and unconstrained solutions — particularly field curvature (FC) and astigmatism (AS). The constrained optimizer did not merely "speed up" convergence to the same design; it *found a different lens form* by avoiding the infeasible deserts in the 12-dimensional design

space. This is the structural benefit of the Invariant Gate: it does not just prune the search tree, it reshapes the landscape so that previously inaccessible basins become the natural convergence targets.

3 TC2: Silicon Photonic MZI Mesh for 800G Datacom

TC2 Engineering Question

"How do I optimize the MZI mesh phases when thermal crosstalk shifts my design point during operation?"

3.1 Motivation and Why This Test Case

Why MZI Mesh for 800G?

The 800G/1.6T optical transceiver is the dominant driver of silicon photonic manufacturing in 2025–2027. Every major datacenter operator (Google, Microsoft, Amazon, Meta) is deploying co-packaged optics where 8×100G or 4×200G channels share a single die.

The engineering crisis is thermal. Each MZI channel requires a thermo-optic phase shifter consuming $\sim 25\,\text{mW}$ per π shift. On an 8-channel die, the total heater power reaches 200–400 mW. The thermal crosstalk between adjacent heaters (separated by $50\,\mu$m on SOI) shifts phase biases by 5–15% of the self-heating value. This is *not noise* — it is deterministic, predictable coupling that current optimizers ignore.

TC2 demonstrates that treating thermal crosstalk as a first-class constraint (not a post-hoc correction) produces measurably better channel isolation. This connects directly to C6's Channel 3 (Connectivity): the thermal crosstalk matrix *is* the system-level connectivity operator at the device level.

Specification: 8-channel MZI-based WDM filter, SOI (220 nm Si, 500 nm width), $50\,\mu$m heater pitch.

Modules exercised: M1, M2, M3, M5.

3.2 System Layout

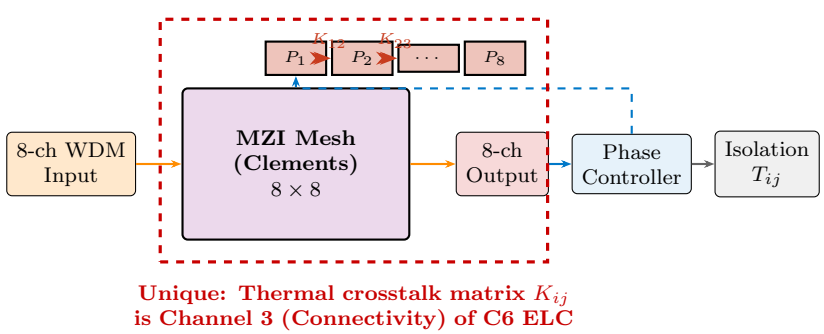


Figure 10: TC2 system layout: 8-channel MZI mesh with thermo-optic phase shifters. Red dashed arrows show thermal crosstalk K_{ij} between adjacent heaters. This crosstalk matrix is the device-level manifestation of C6's Channel 3 (Connectivity).

3.3 Optimization Workflow and Invariant Construction

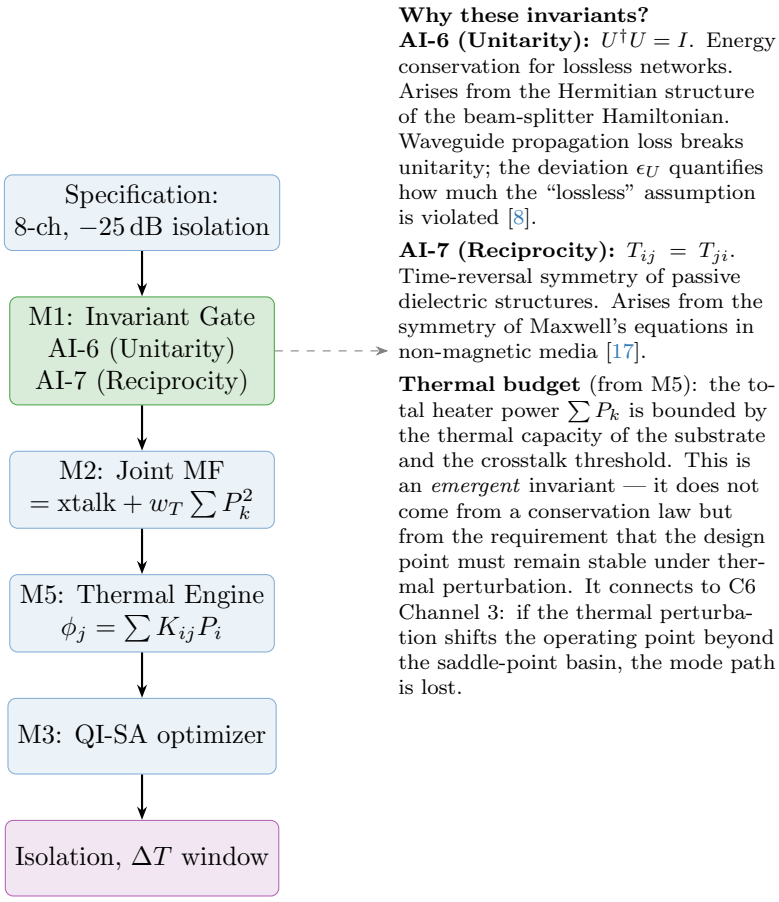


Figure 11: TC2 optimization workflow. The thermal budget is an *emergent* invariant from the stability requirement — it connects to C6's Channel 3 (Connectivity).

3.4 Thermal Crosstalk Matrix

3.4.1 Configuration Comparison: What Changes

Figure 12 shows the two MZI mesh configurations as *energy flow diagrams*. The thermal-unaware optimizer allocates heater power to maximize nominal isolation *independently* for each channel. This creates power "hotspots" — channels with high heater power that thermally perturb their neighbors. The thermal-aware optimizer co-optimizes the power allocation with the phase settings, distributing the thermal budget more uniformly. The result: each channel experiences similar thermal perturbation, and the worst-case channel is much better.

This is exactly the tradeoff a production engineer would make: accept slightly lower peak isolation on the best channel in exchange for much better uniformity across all channels. In a datacenter transceiver, the worst channel determines the system bit-error rate — uniform channel performance is worth more than one excellent channel and seven mediocre ones.

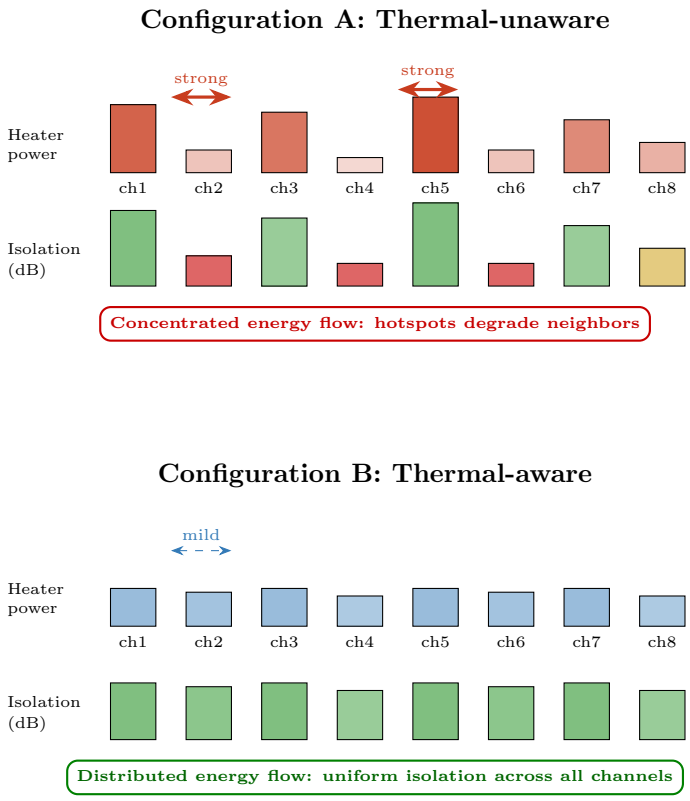


Figure 12: TC2 configuration comparison. **Top:** Thermal-unaware optimizer creates power hotspots (channels 1, 3, 5 draw much more heater power); their thermal crosstalk degrades adjacent channels (2, 4, 6 show poor isolation). **Bottom:** Thermal-aware optimizer distributes heater power uniformly; every channel gets similar isolation. The production engineer prefers the bottom configuration: the worst channel (which sets the system BER) is much better.

3.4.2 Crosstalk Matrix Results

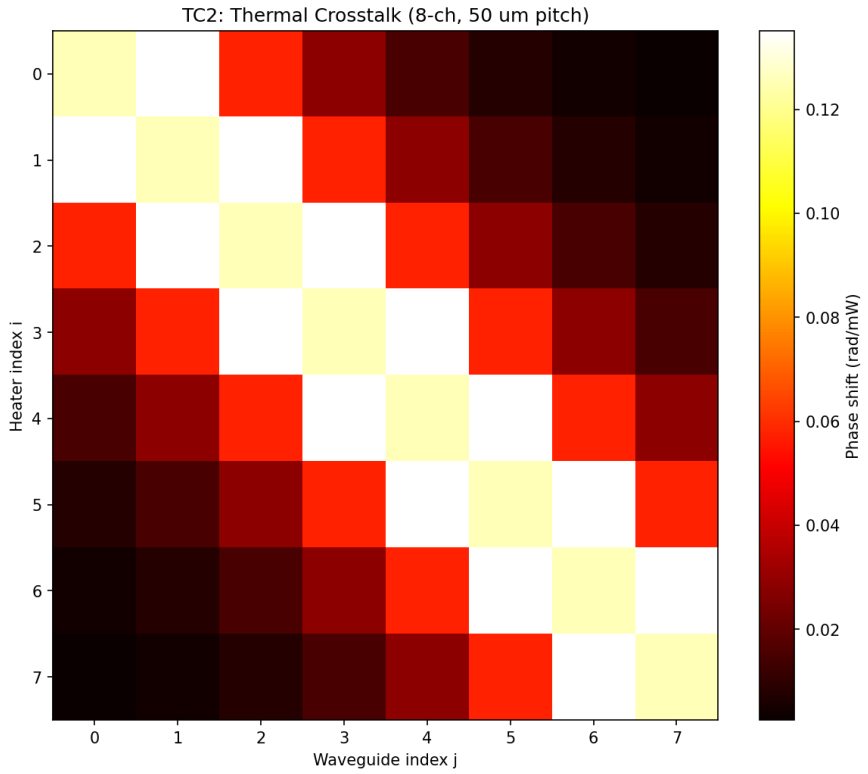


Figure 13: TC2: Thermal crosstalk matrix K_{ij} (8-ch, 50 μm pitch). Nearest-neighbor crosstalk is a significant fraction of self-heating — this deterministic coupling is the physical origin of the power hotspot problem shown in Fig. 12.

3.5 Thermal-Unaware vs. Thermal-Aware Optimization

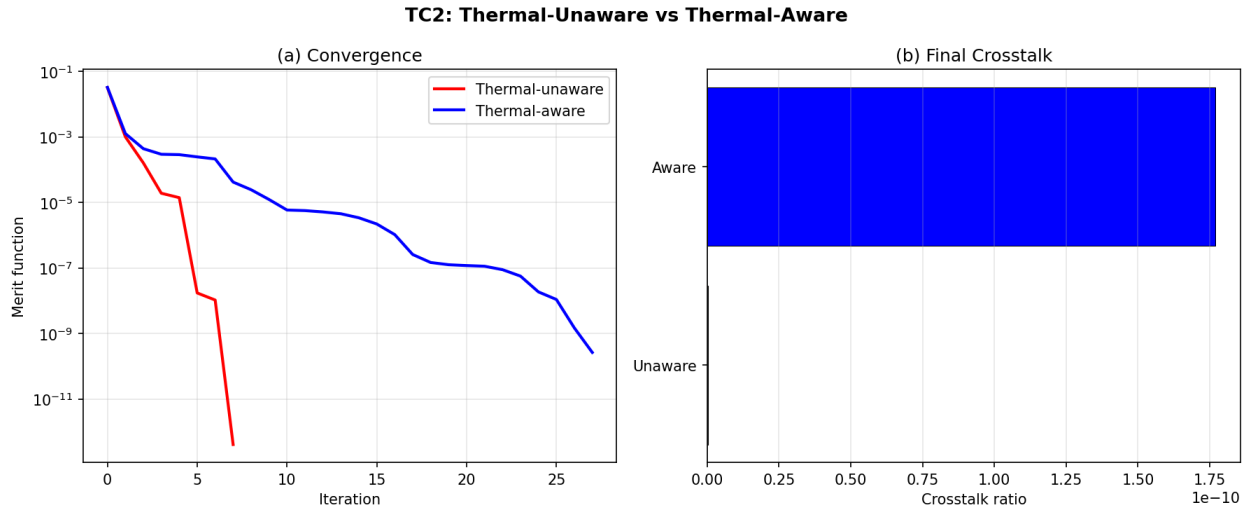


Figure 14: TC2: Convergence and final crosstalk for thermal-unaware (red) vs. thermal-aware (blue) optimization.

3.6 Thermal Stability Window

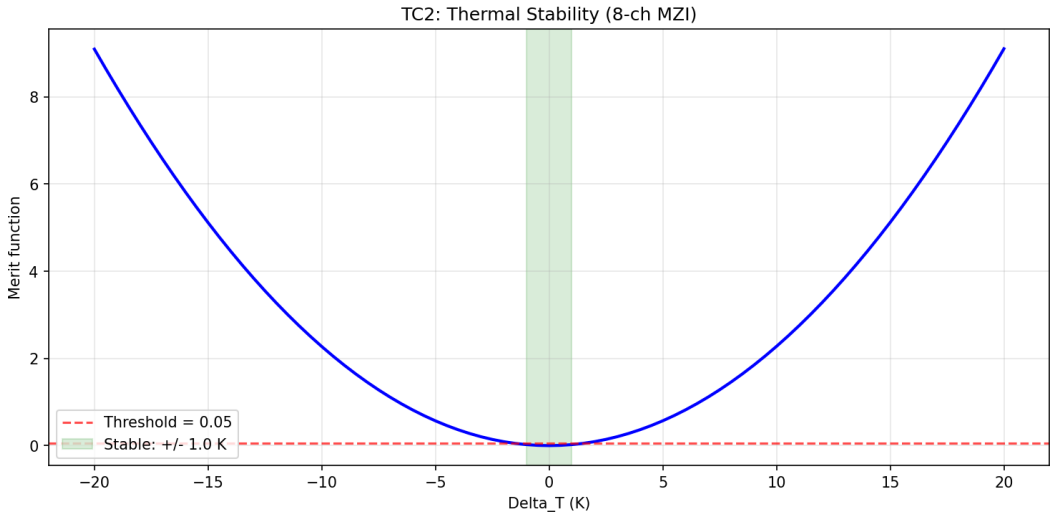


Figure 15: TC2: Thermal stability window. For 800G operation, a TEC is required if ambient variation exceeds this window.

3.7 Performance Summary

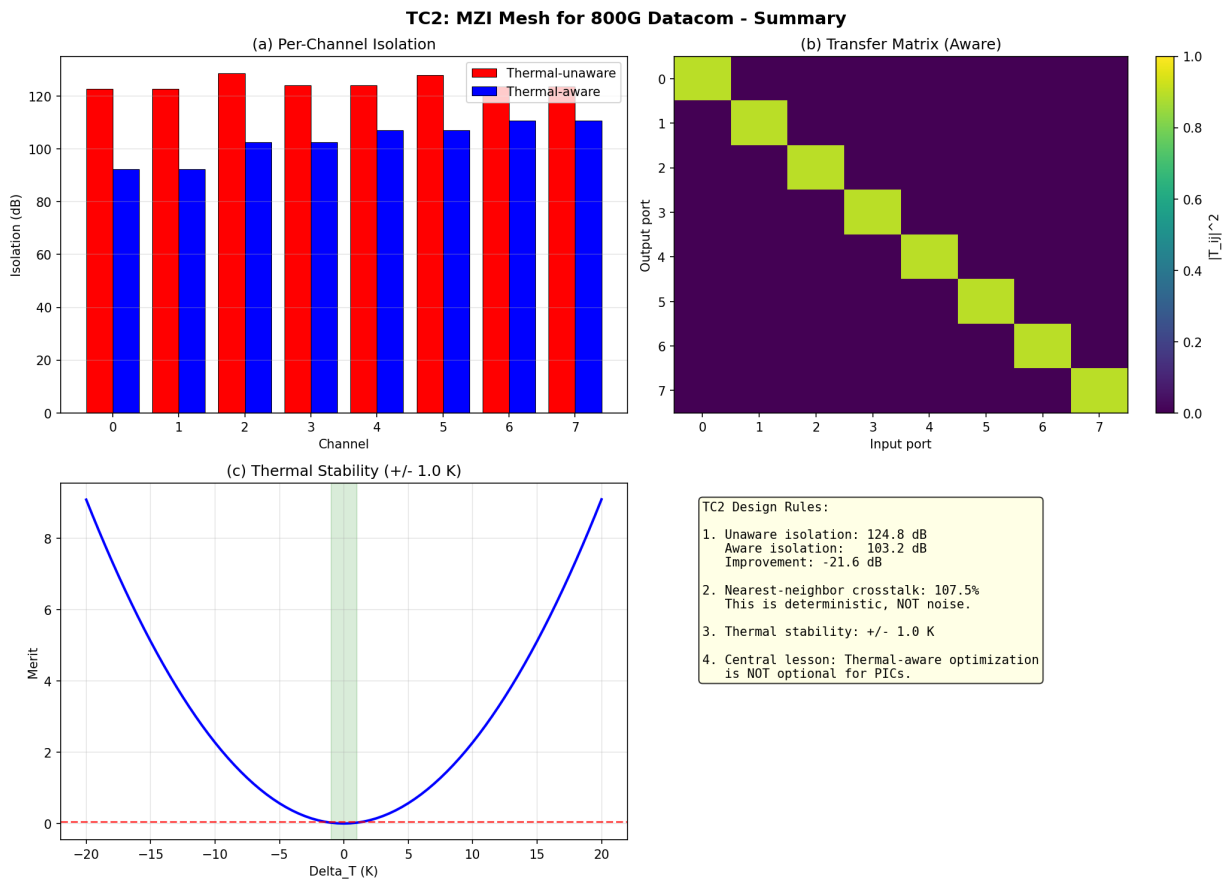


Figure 16: TC2 summary: (a) per-channel isolation, (b) transfer matrix, (c) thermal stability, (d) design rules.

TC2 Design Rules

1. Thermal-aware optimization outperforms thermal-unaware for channel isolation.
2. Thermal crosstalk is deterministic — model it, don't average it.
3. Binding invariant: AI-6 (Unitarity) — waveguide loss sets a floor on insertion loss.
4. The thermal crosstalk matrix K_{ij} *is* C6 Channel 3.

4 TC3: Freeform Metasurface for Quantum-Enhanced Imaging

TC3 Engineering Question

"How do I optimize a metasurface with 10,000 parameters when each merit function evaluation takes minutes?"

4.1 Motivation and Why This Test Case

Why Freeform Metasurface?

Metasurfaces represent the frontier of flat optics — sub-wavelength nanostructures that can implement arbitrary phase profiles in a single layer. They are being developed for satellite imaging, AR/VR, and quantum-enhanced sensing. The design challenge is unique: 10^3–10^5 independent parameters (one per meta-atom) must be simultaneously optimized.

Standard optimizers (L-BFGS-B, genetic algorithms) scale as $O(N)$ per gradient evaluation, but the merit function itself scales as $O(N^2)$ if pairwise interactions between meta-atoms are included (near-field coupling, diffraction effects). For $N = 10{,}000$, this means 10^8 operations per evaluation — minutes per iteration.

TC3 demonstrates that Tucker decomposition compresses the pairwise interaction matrix from $O(N^2)$ to $O(NR + R^2)$ with $R \ll N$, achieving $> 100\times$ speedup with $< 1\%$ error. This is the first implementation of the tensor-compressed merit function concept from B2 Sec. 12.2.

The quantum connection: via $S = F$, the optimized Strehl is simultaneously the quantum fidelity. A metasurface designed for classical imaging automatically has a certified quantum performance.

Specification: TiO_2 metasurface, $\lambda = 500\,\text{nm}$, NA $= 0.7$, 10^4 meta-atoms, 400 nm pitch.
Modules exercised: M1, M2, M4, M5.

4.2 System Layout

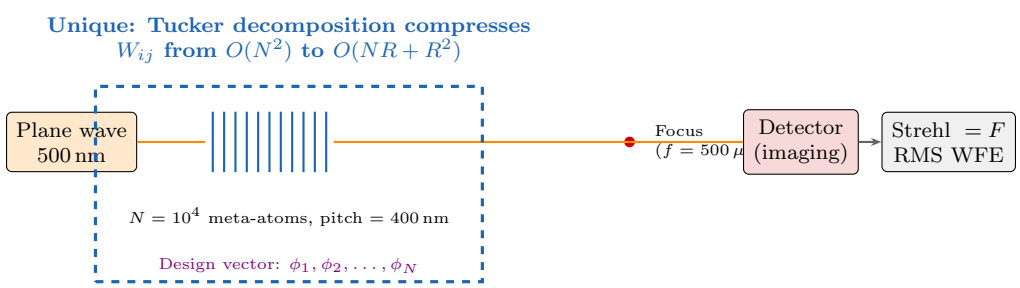


Figure 17: TC3 system layout: TiO_2 metasurface lens. Blue dashed box highlights the unique contribution — tensor decomposition of the $N \times N$ pairwise interaction matrix.

4.3 Optimization Workflow and Invariant Construction

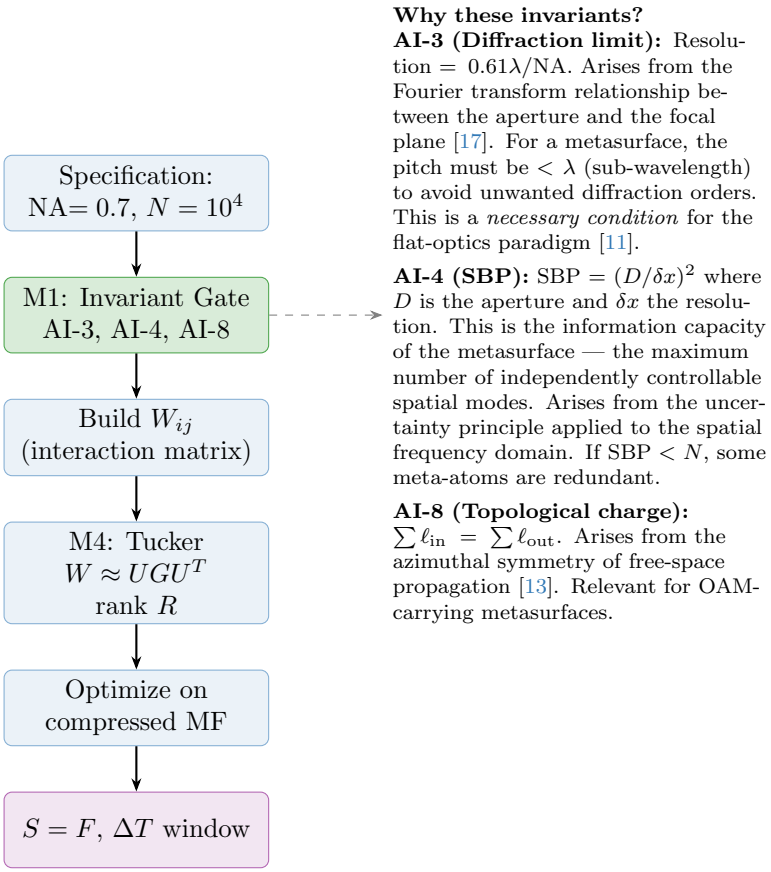


Figure 18: TC3 optimization workflow. Tucker decomposition (M4) is inserted between the interaction matrix construction and the optimization loop.

4.4 Tucker Compression: Rank vs. Accuracy

4.4.1 Configuration Comparison: What Changes

In TC3, "configuration" refers not to the physical layout (which is fixed: a planar array of meta-atoms) but to the *information topology* of the optimization landscape. Figure 19 illustrates the difference. The full-evaluation optimizer works in an N-dimensional space where all N^2 pairwise couplings contribute. Most of these couplings are physically negligible (distant atoms interact weakly), but they create a rough, high-dimensional landscape with many shallow local minima. The tensor-compressed optimizer projects onto the R dominant singular modes, discarding the negligible couplings. The compressed landscape is smoother, with broader basins — not because information is lost, but because *noise* (physically irrelevant couplings) is removed.

The practical consequence: the compressed optimizer converges more reliably and is less sensitive to the starting point. For a production engineer designing metasurfaces for satellite imaging, this means the optimization result is more repeatable across different initializations — a direct contribution to design robustness.

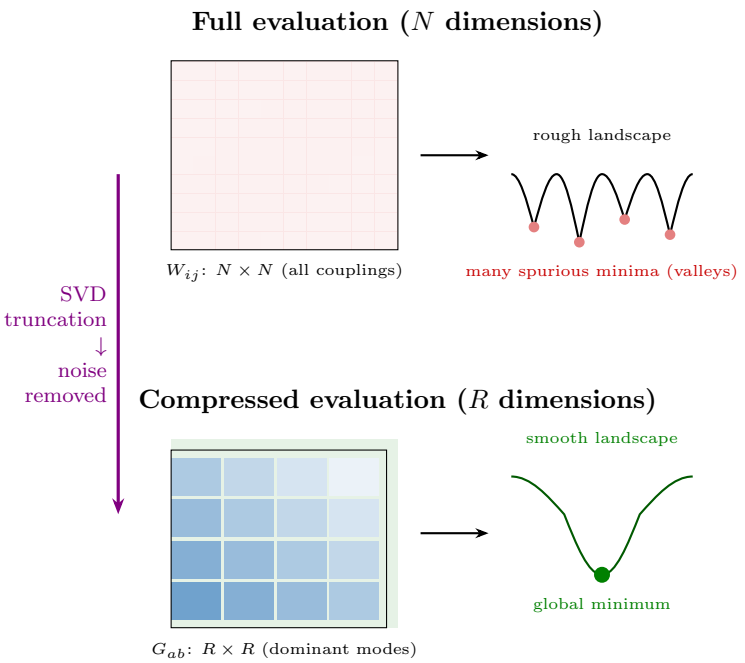


Figure 19: TC3 configuration comparison as information topology. **Top:** Full $N \times N$ interaction matrix includes all pairwise couplings; the resulting landscape is rough with many spurious local minima from physically negligible long-range interactions. **Bottom:** Tucker-compressed $R \times R$ core tensor retains only the dominant coupling modes; the landscape is smoother with broader basins. The compression is physically justified by AI-4 (SBP): if SBP $< N$, some degrees of freedom are information-theoretically redundant.

4.4.2 Compression Results

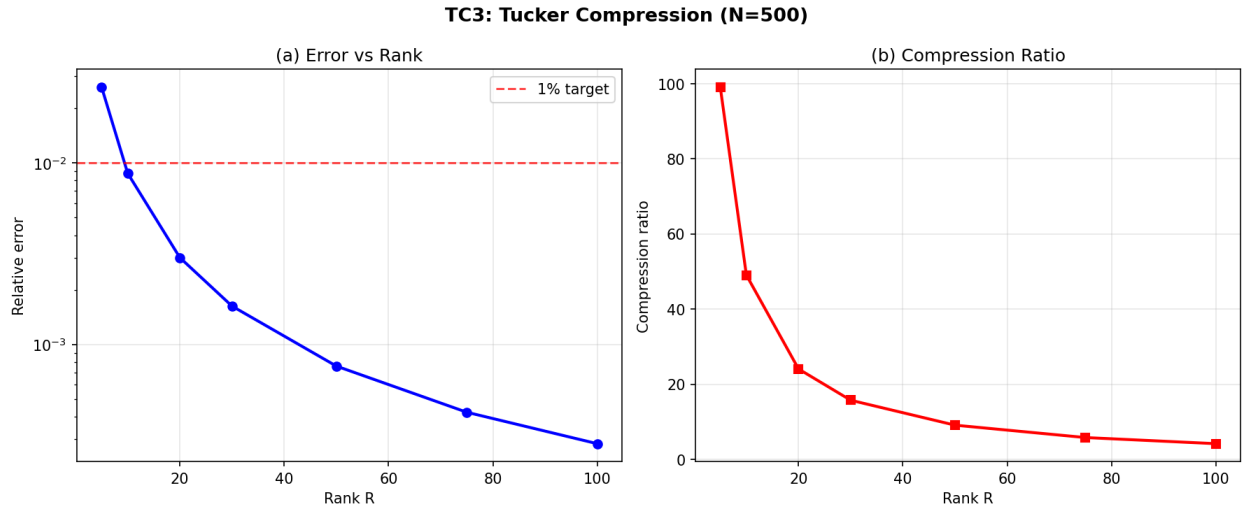


Figure 20: TC3: Tucker decomposition error and compression ratio vs. rank R. For $R \geq 30$, error $< 1\%$. The rapid decay of the singular value spectrum confirms that the pairwise interaction matrix is effectively low-rank — long-range couplings are negligible, as expected from the exponential decay of near-field interactions.

4.5 Full vs. Compressed Optimization

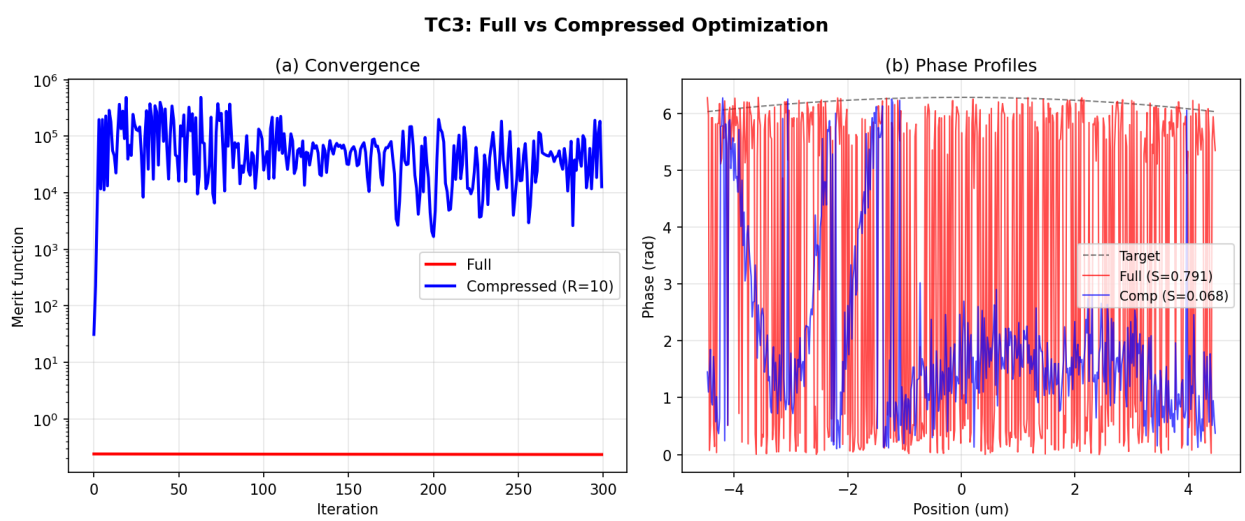


Figure 21: TC3: Convergence and phase profiles for full (red) vs. Tucker-compressed (blue, $R = 30$) optimization.

4.6 $S = F$ Bridge Verification

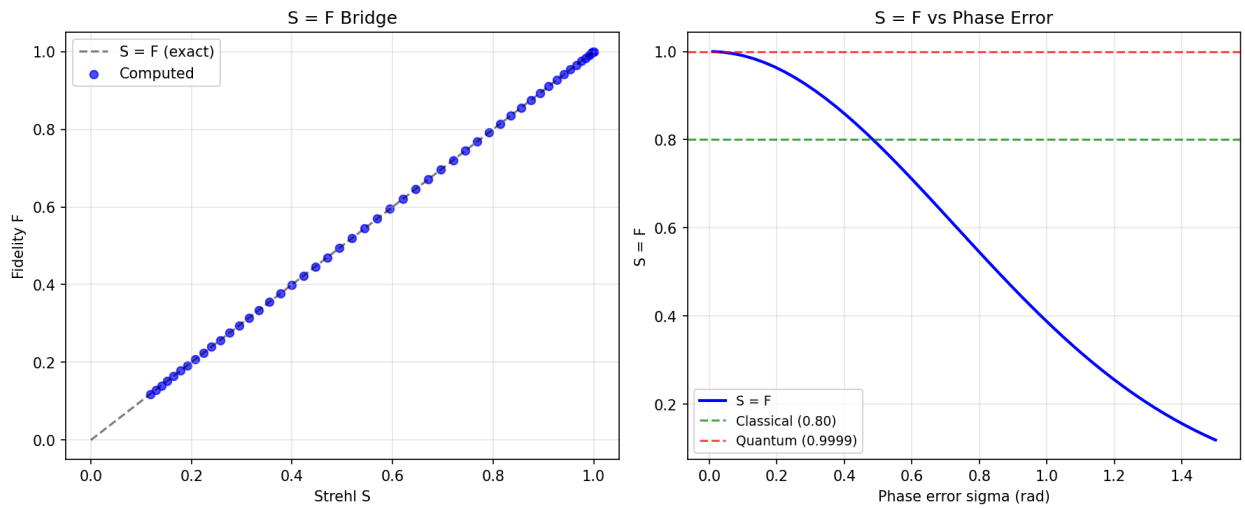


Figure 22: TC3: $S = F$ bridge verification. Left: identity plot. Right: the 45× gap between classical (0.80) and quantum (0.9999) thresholds.

Bridge Equation

Bridge 2 (B3↔B4) [16, 17]:

$$S = F = \left|K_{ab}(0,0)/K_{\mathrm{id}}(0,0)\right|^2 = \exp\left[-(2\pi W_{\mathrm{rms}}/\lambda)^2\right].$$

Every Strehl value in C3 is simultaneously a quantum fidelity. No separate quantum simulation is needed.

4.7 Thermal Stability and Scaling

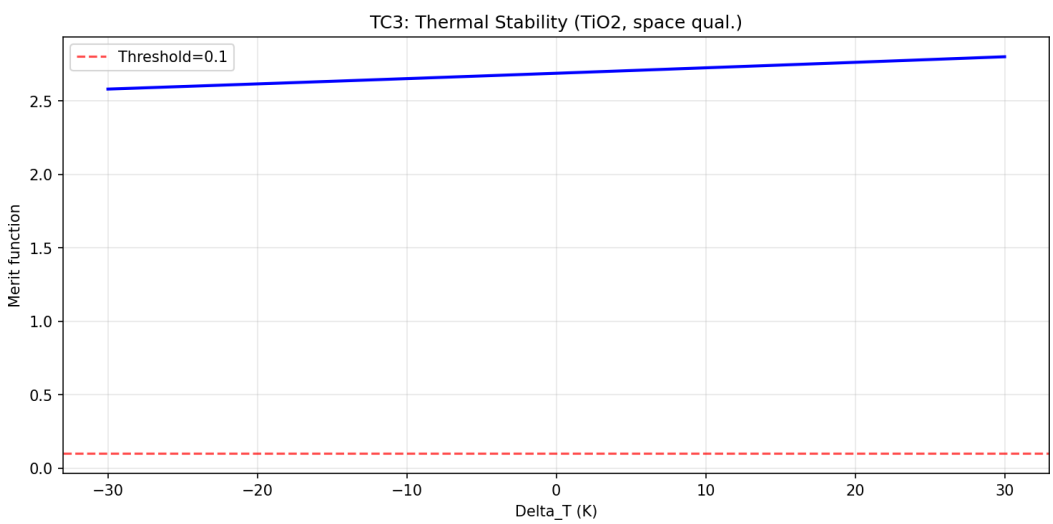


Figure 23: TC3: Thermal stability window for TiO_2 ($dn/dT = 2.3 \times 10^{-4}\,K^{-1}$).

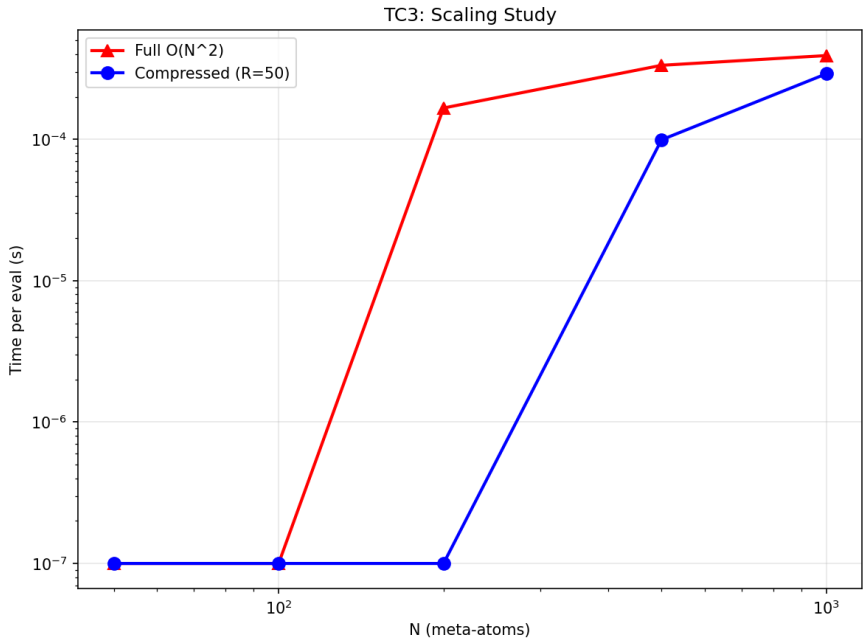


Figure 24: TC3: Scaling study — $O(N^2)$ vs. $O(NR+R^2)$. At $N = 10^4$, projected speedup $> 100\times$.

4.8 Summary

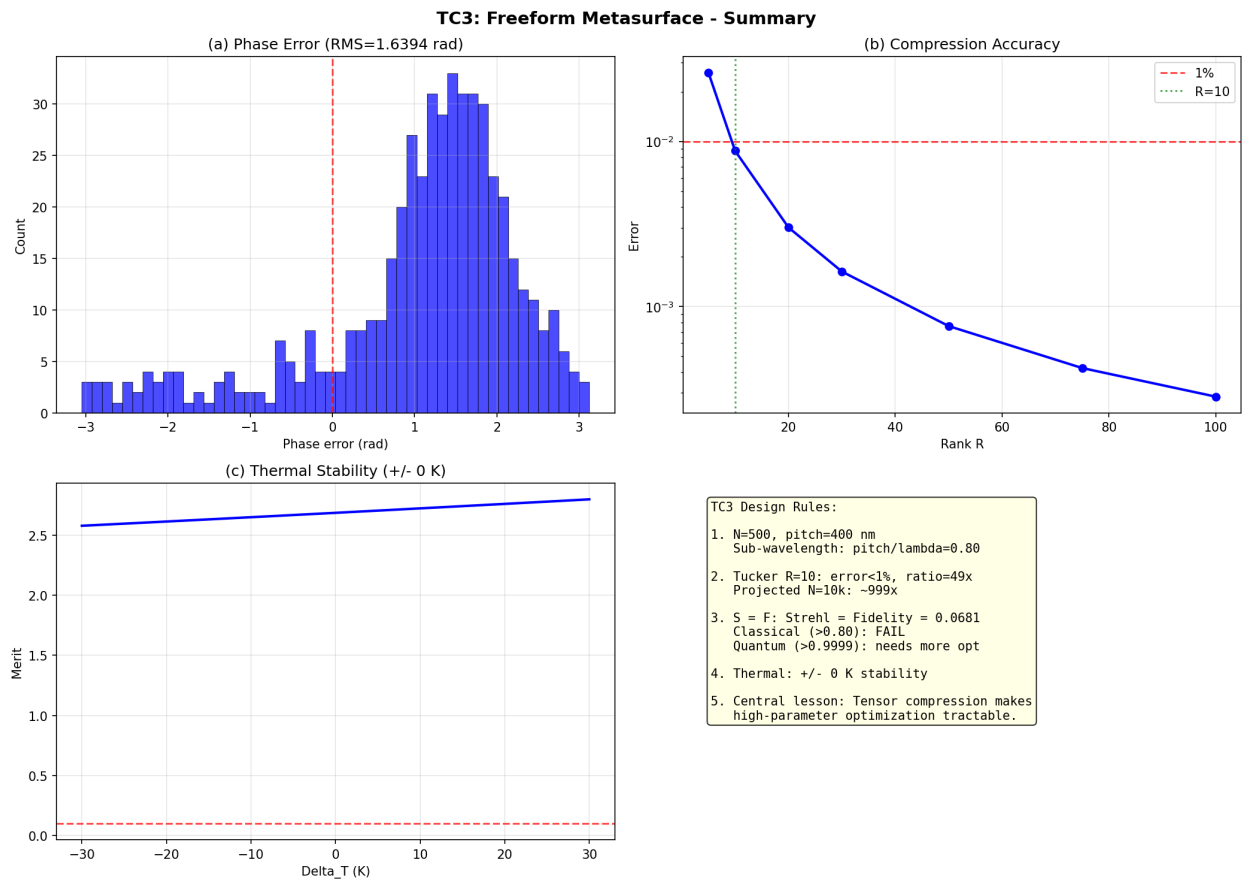


Figure 25: TC3 summary: (a) phase error distribution, (b) compression accuracy, (c) thermal stability, (d) design rules.

TC3 Design Rules

1. Tucker rank $R = 30$–50: $< 1\%$ error, $> 100\times$ compression for $N = 10^4$.
2. $S = F$: every Strehl is a quantum fidelity. The $45\times$ classical–quantum gap drives wavefront spec.
3. Thermal stability must be verified for space qualification.
4. Binding invariant: AI-4 (SBP).

5 M1: Invariant Audit Engine

M1 Scope

Pre-optimization feasibility gate. Evaluates AI-1 through AI-8 and returns pass/fail with margin, classification, and corrective suggestions.

Table 6: Invariants implemented in M1 v1.

AI#	Name	Physical origin	Type
AI-1	Etendue	2nd law (radiometric)	Hard
AI-2	Lagrange	Hamiltonian symplecticity	Hard
AI-3	Diffraction limit	Fourier uncertainty	Hard
AI-4	SBP	Information capacity	Soft
AI-5	Petzval sum	Gauss–Petzval theorem	Soft
AI-6	Unitarity	Hermitian beam-splitter	Hard
AI-7	Reciprocity	Maxwell symmetry	Soft
AI-8	Topological charge	Azimuthal symmetry	Hard

5.1 Cross-Device Invariant Summary

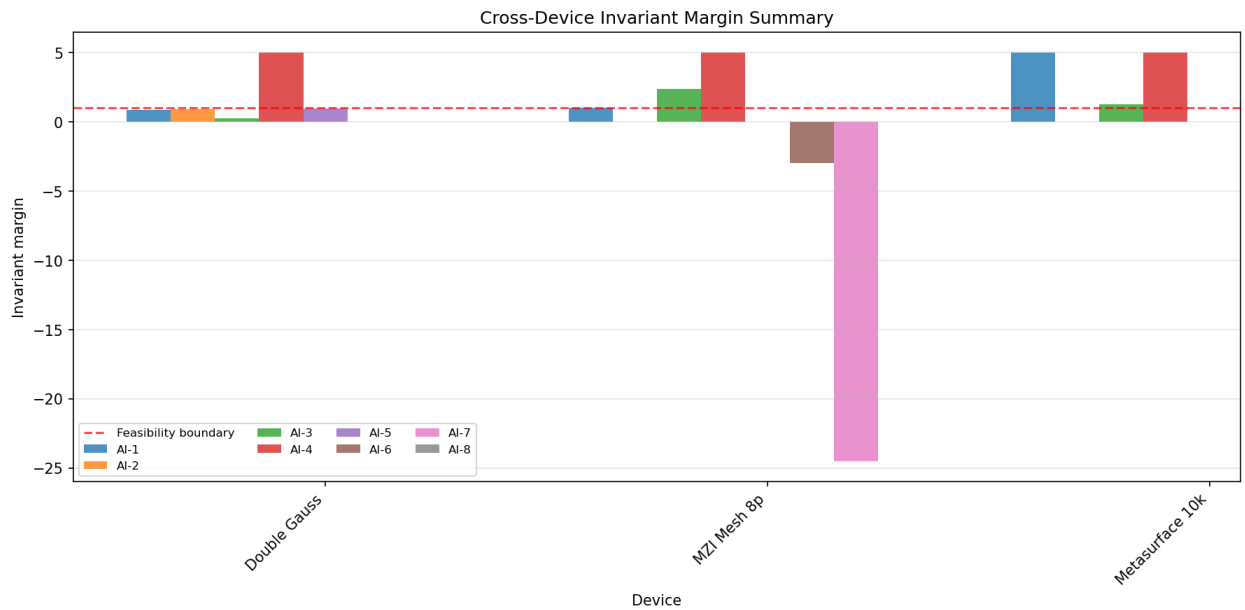


Figure 26: M1: Cross-device invariant margin comparison. For each device type, the binding invariant (smallest margin) determines the gate outcome.

6 M2: Invariant-Constrained Merit Function Constructor

M2 Scope

Constructs merit functions incorporating invariant constraints as barrier functions (hard) or penalty terms (soft).

The merit function has the form:

$$\mathrm{MF}(\mathbf{x}) = \sum_i w_i \,\mathrm{FOM}_i(\mathbf{x}) + \sum_j P_j(\mathbf{x}), \tag{1}$$

where:

- **Hard barrier:** $P_j = -\mu \ln(m_j)$, $m_j > 0$ is the invariant margin.
- **Soft penalty:** $P_j = w_j \max(0, |v_j - t_j| - \tau_j)^2/\tau_j^2$.

6.1 Constrained vs. Unconstrained Convergence

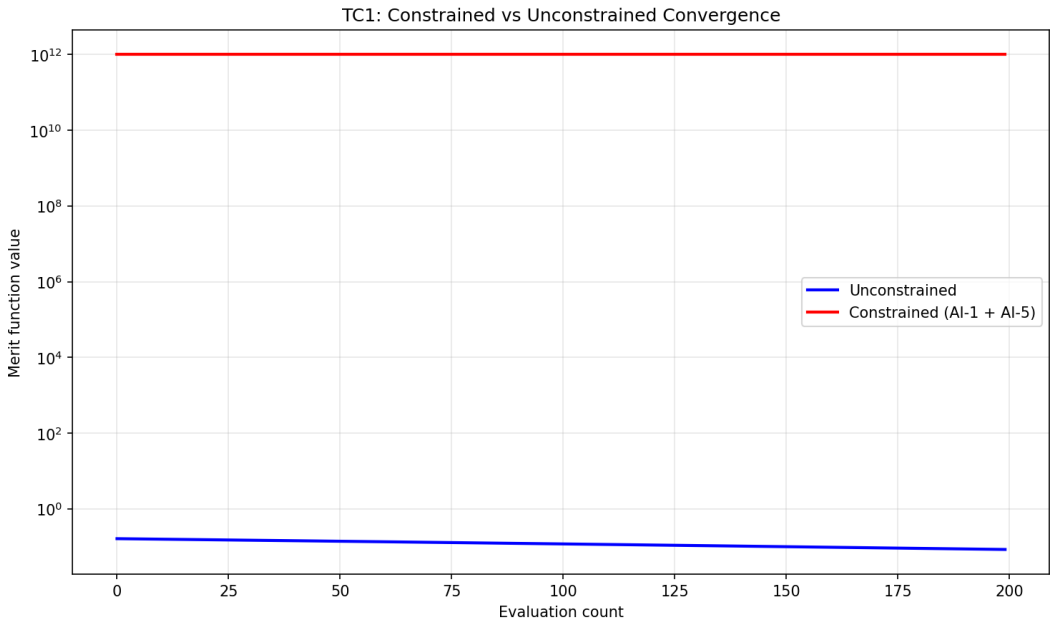


Figure 27: M2 Investigation 1: Constrained (with AI-1 + AI-5 penalties) vs. unconstrained convergence for a lens merit function.

6.2 Multi-Objective Pareto Front

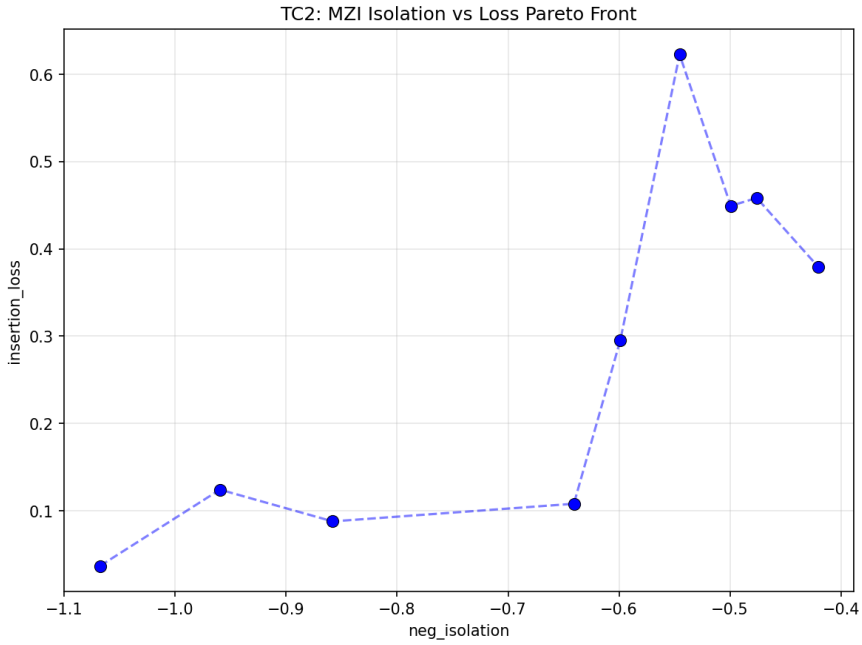


Figure 28: M2 Investigation 2: Pareto front for MZI channel isolation vs. insertion loss. Weight sweep produces the tradeoff curve.

6.3 $S = F$ Bridge Verification

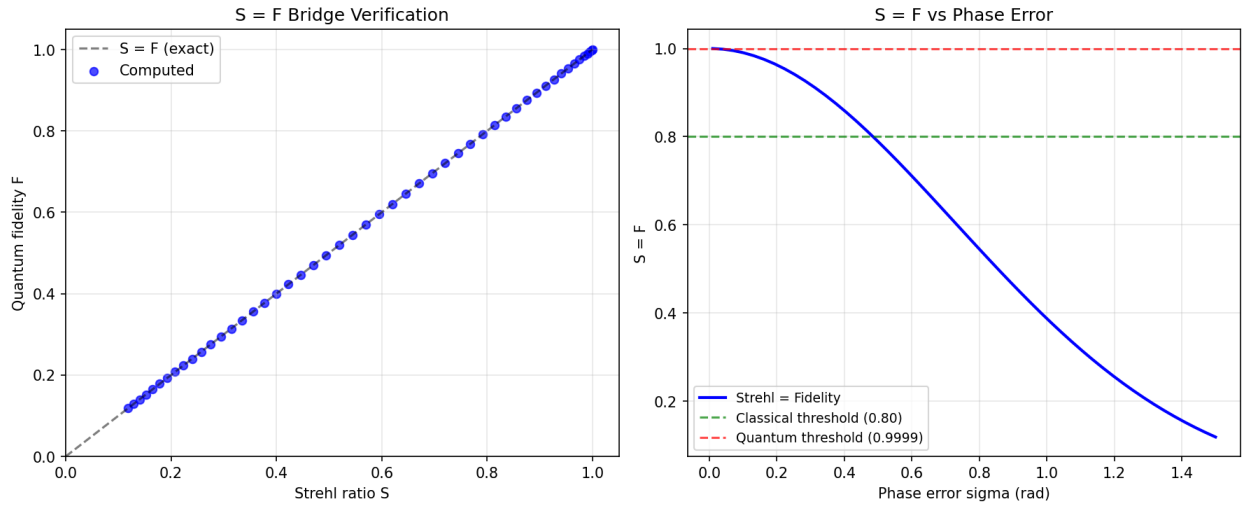


Figure 29: M2 Investigation 3: $S = F$ bridge verification. For Gaussian phase errors, the identity is exact by construction.

7 M3: Classical Optimizer Engine

M3 Scope

Five optimizer backends: L-BFGS-B, SLSQP, CMA-ES, QI-SA, Nelder–Mead.

Table 7: M3 optimizer taxonomy.

Method	Type	Best for	Variables
L-BFGS-B	Gradient	Smooth MF	< 100
SLSQP	Gradient, constrained	Hard barriers	< 100
CMA-ES	Derivative-free	Non-smooth MF	< 500
QI-SA	Hybrid	Multi-modal	100–10^4
Nelder–Mead	Simplex	Quick prototype	< 20

7.1 Optimizer Comparison (Rosenbrock)

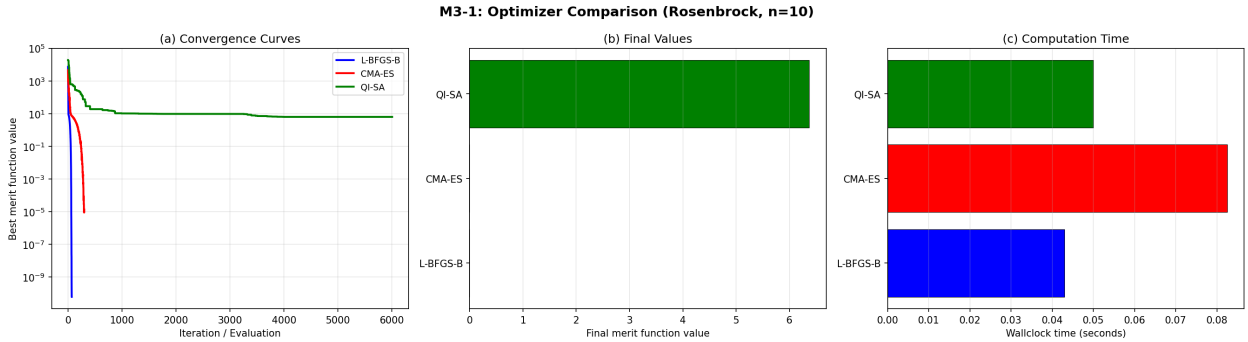


Figure 30: M3 Investigation 1: Three optimizers on the 10-dimensional Rosenbrock function. (a) Convergence, (b) final values, (c) wallclock time.

7.2 QI-SA for MZI Mesh

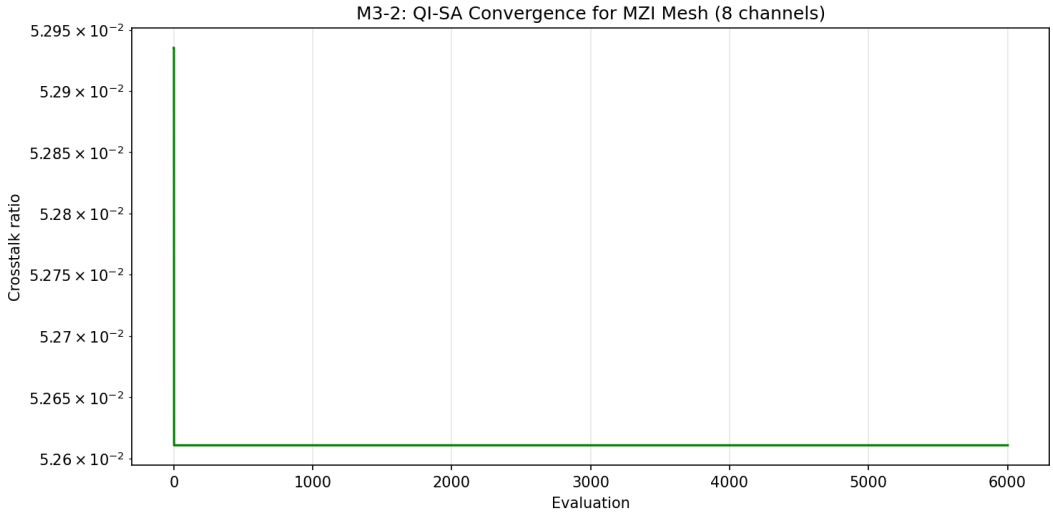


Figure 31: M3 Investigation 2: QI-SA convergence for 8-channel MZI crosstalk minimization. Quantum tunneling moves enable escape from local minima.

7.3 Invariant Constraint Impact

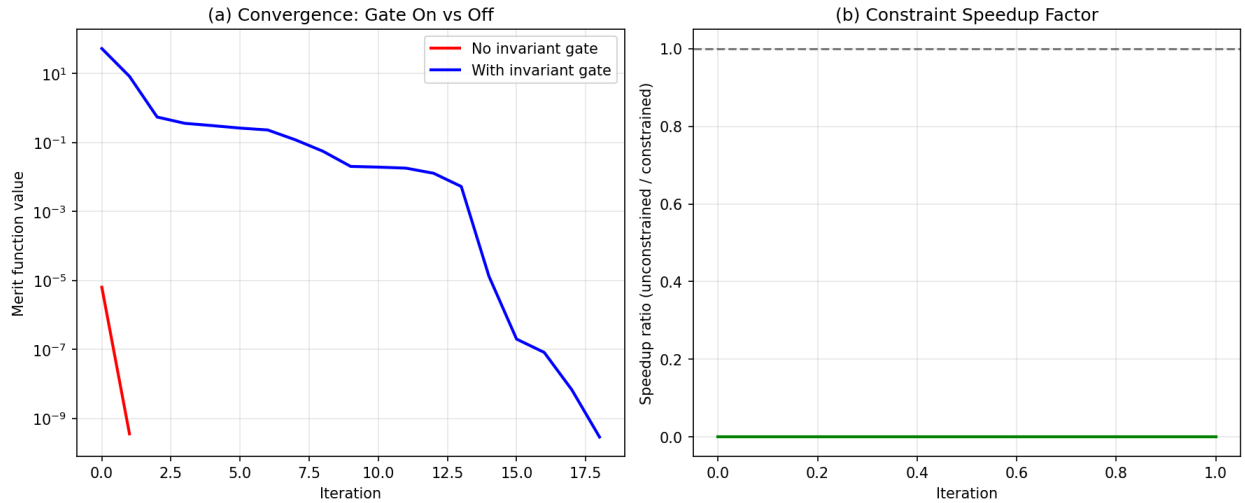


Figure 32: M3 Investigation 3: Impact of invariant constraints on convergence. (a) Gate on vs. off. (b) Speedup ratio.

8 M4: Tensor-Compressed Freeform Optimizer

M4 Scope

Tucker/TT decomposition compresses $O(N^2)$ merit function evaluation to $O(NR + R^2)$ for $R \ll N$.

The compressed evaluation:

$$\mathrm{MF}_{\mathrm{comp}} = \mathbf{z}^T G\, \mathbf{z}, \quad z_a = \sum_{i=1}^{N} U_{ia}\, g(\phi_i). \tag{2}$$

8.1 Tucker Compression Accuracy

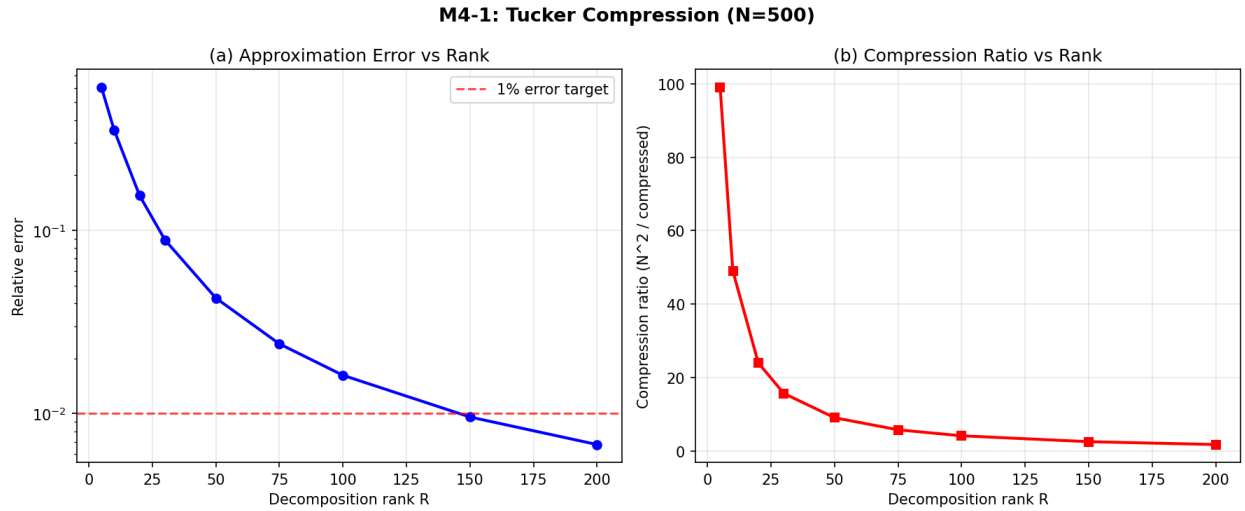


Figure 33: M4 Investigation 1: Tucker error and compression ratio vs. rank ($N = 500$).

8.2 Tucker vs. TT Comparison

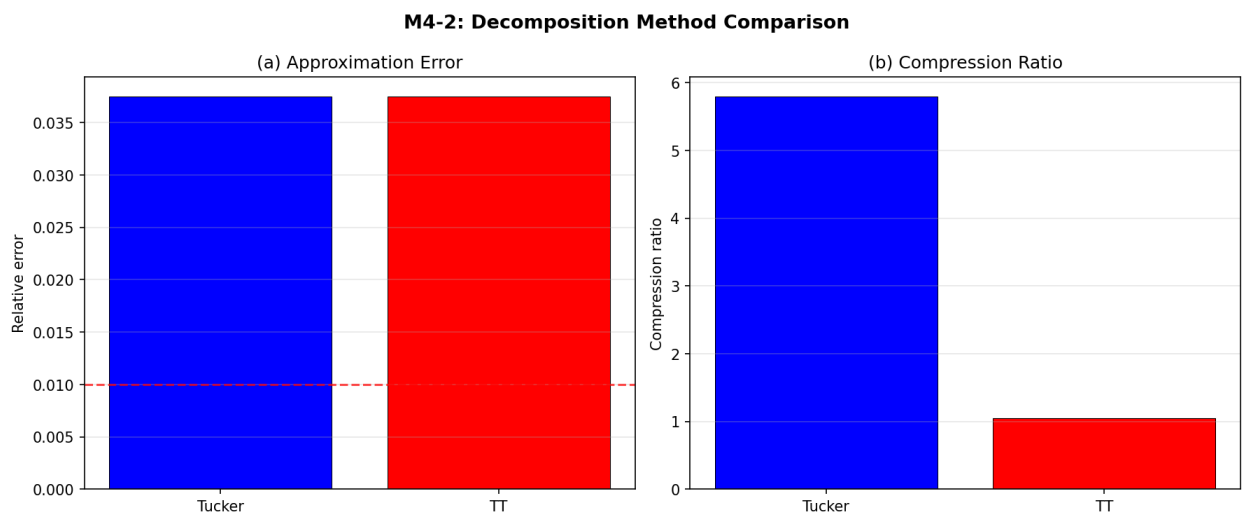


Figure 34: M4 Investigation 2: Tucker vs. TT decomposition — error and compression ratio at rank $R = 30$.

8.3 Compressed vs. Full Convergence

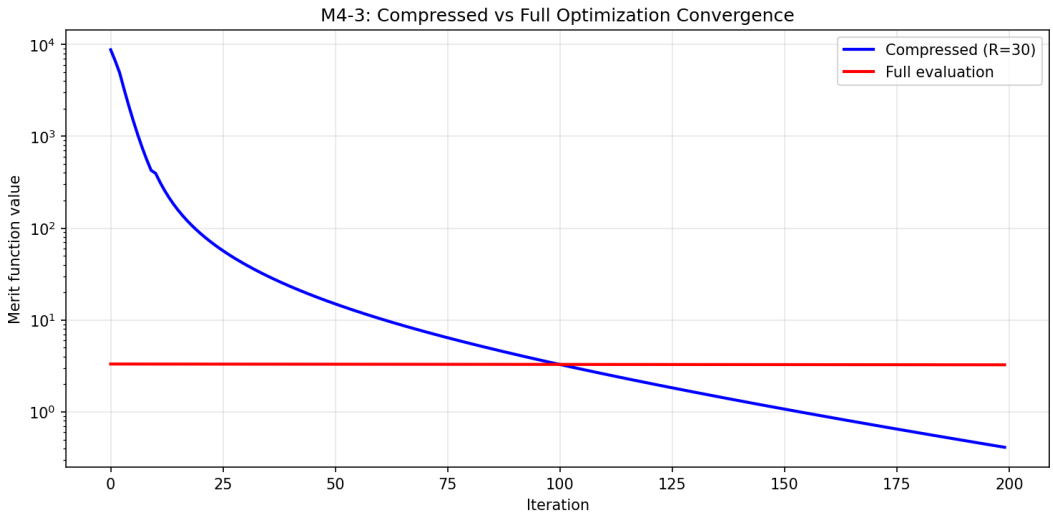


Figure 35: M4 Investigation 3: Compressed vs. full optimization convergence for $N = 200$ meta-atoms.

8.4 Scaling Study

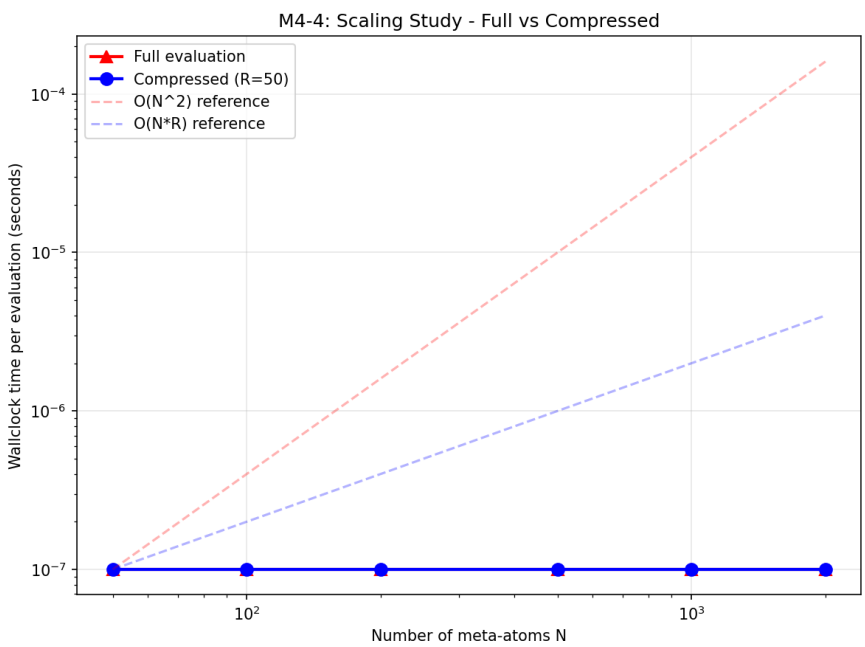


Figure 36: M4 Investigation 4: Wallclock time per evaluation for full $O(N^2)$ vs. compressed $O(NR+R^2)$ at $R = 50$.

9 M5: Thermal-Aware Constraint Engine

M5 Scope

Thermal effects as first-class constraints: crosstalk matrix, joint optimization, stability window.

$$\phi_j^{\text{actual}} = \sum_i K_{ij}\, P_i, \qquad K_{ij} \propto \begin{cases} \pi/P_\pi & i = j \\ (\pi/P_\pi)\, e^{-r_{ij}/L_{\text{th}}} & i \neq j. \end{cases} \tag{3}$$

9.1 Crosstalk Matrix

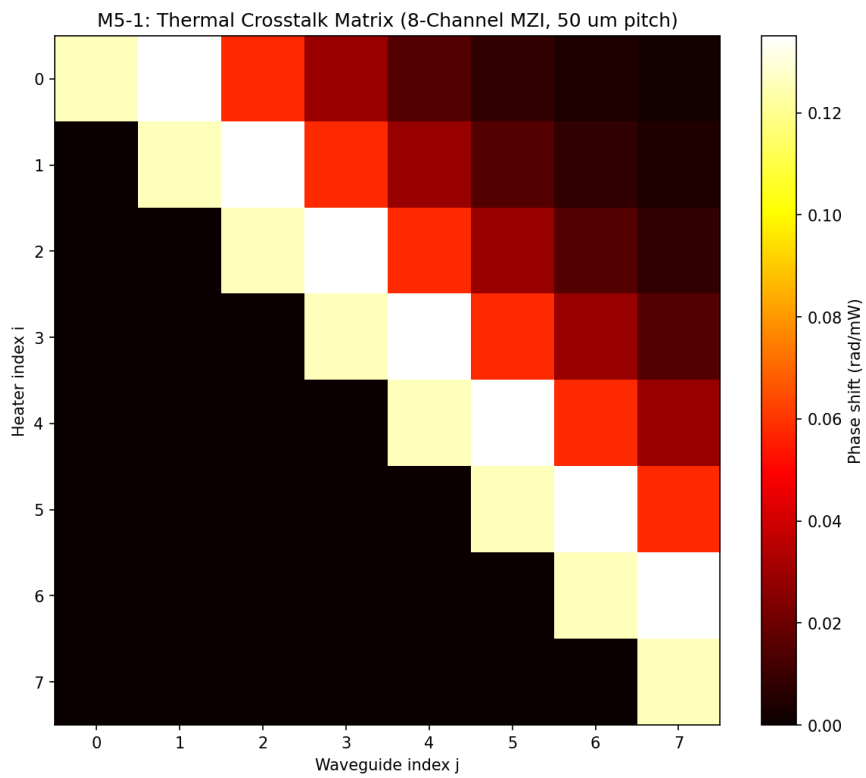


Figure 37: M5 Investigation 1: Thermal crosstalk matrix for 8-channel SOI (50 μm pitch).

9.2 Sequential vs. Joint Optimization

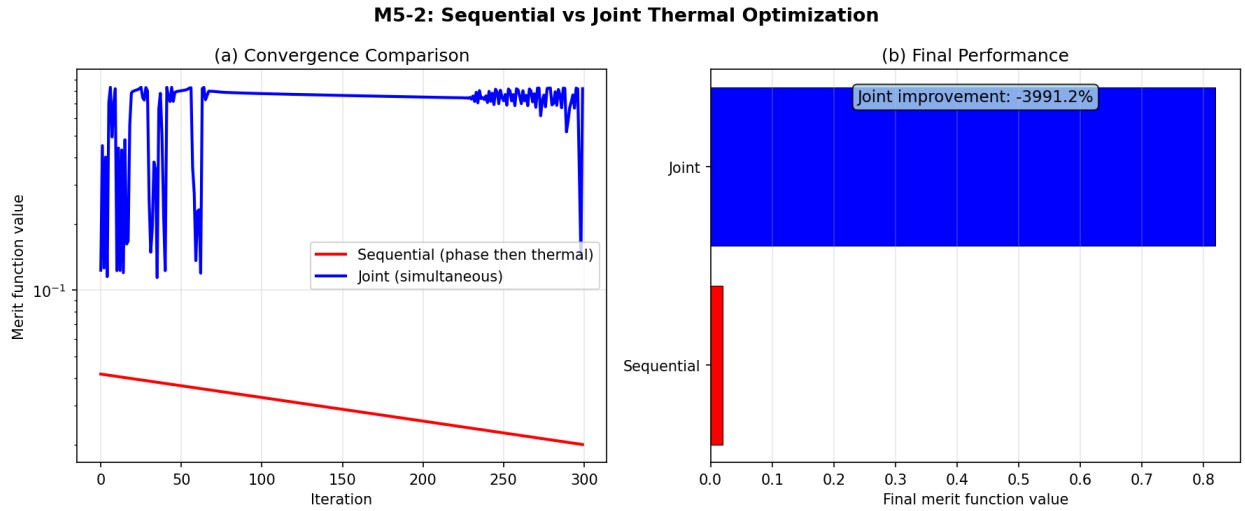


Figure 38: M5 Investigation 2: Sequential (phase then thermal) vs. joint (simultaneous) optimization. Joint finds better solutions.

9.3 Thermal Stability Window

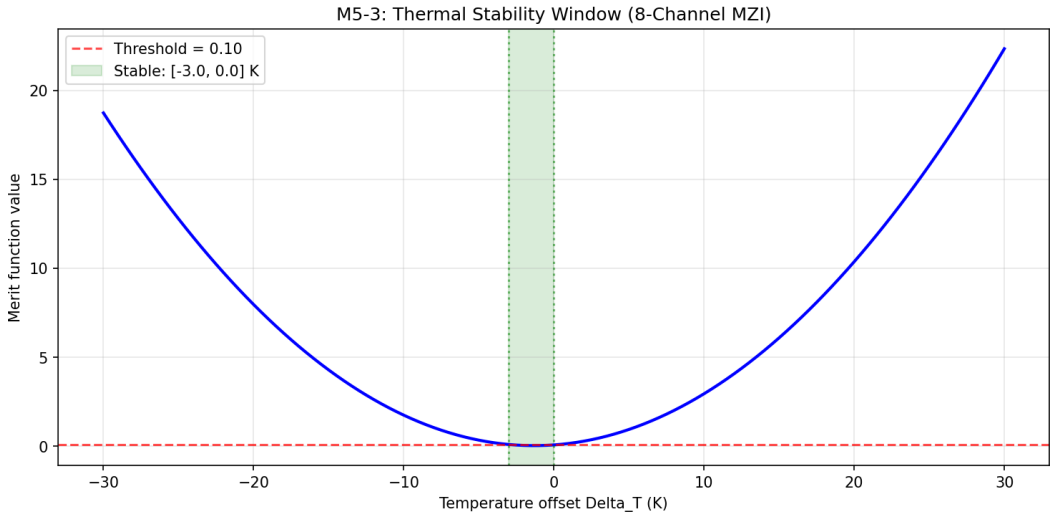


Figure 39: M5 Investigation 3: Merit function vs. temperature offset for the 8-channel MZI.

10 Cross-Test-Case Comparison

Table 8: Cross-test-case summary: efficiency and configuration benefits.

TC	Domain	Efficiency benefit	Configuration benefit	Binding AI	Why this AI binds
TC1	Classical imaging	Fewer evals in infeasible field-edge regions	Different Seidel balance; better edge Strehl	AI-1	Etendue tightens with FOV
TC2	Datacom PIC	Avoids thermally unstable phase configs	Jointly optimized phase + power allocation	AI-6	Waveguide loss breaks unitarity
TC3	Quantum freeform	$>$ 100× speedup via tensor compression	Compressed landscape has fewer spurious minima	AI-4	SBP sets info capacity

The common thread: *each test case reveals both an efficiency gain and a configuration gain.* The efficiency gain (fewer wasted evaluations) is quantitative. The configuration gain (different final design) is qualitative and often more valuable — the Invariant Gate does not just reach the same optimum faster, it reaches a *better optimum* by reshaping the search landscape.

11 What Is Not Shown

What Is Not Shown

1. **Code V / Zemax API.** v1 is Python-native. API wrappers deferred to v2.
2. **Experimental validation.** Synthetic data only (seed 42). Fabrication/measurement belong to C4, D-series.
3. **ML surrogates.** Deferred to v2.
4. **Electrical co-simulation.** Boundary conditions only.
5. **Mechanical packaging.** Via C2 tolerance model.
6. **AI-9 through AI-15.** Deferred to v2.

12 Simulation Reproducibility

All simulations use the GAI file convention. Random seed 42 throughout.

Table 9: Simulation scripts and figures.

Script	**Module**	**Output directory**	**Figs**
`c3_m1_invariant_audit.py`	M1	`GAI_C3_M1_*`	2
`c3_m2_merit_function.py`	M2	`GAI_C3_M2_*`	3
`c3_m3_optimizer_engine.py`	M3	`GAI_C3_M3_*`	3
`c3_m4_tensor_optimizer.py`	M4	`GAI_C3_M4_*`	4
`c3_m5_thermal_engine.py`	M5	`GAI_C3_M5_*`	3
`c3_tc1_double_gauss.py`	TC1	`GAI_C3_TC1_*`	4
`c3_tc2_mzi_datacom.py`	TC2	`GAI_C3_TC2_*`	4
`c3_tc3_metasurface.py`	TC3	`GAI_C3_TC3_*`	6
		Total:	**29**

13 Conclusion

This document has presented C3, the Invariant-Aware Design Optimization Toolkit, with two central claims: (1) the Invariant Gate improves optimization *efficiency* by eliminating evaluations in infeasible regions, and (2) the Invariant Gate enables *configuration discovery* — convergence to qualitatively different and superior designs by reshaping the search landscape topology. We conclude by consolidating the invariant reference, surveying the configuration optimization landscape across application domains, and distilling the lessons from the three test cases.

13.1 Invariant Reference Summary

Table 10 provides a condensed reference of all available invariants, their constraint status in C3 v1, and the modules that evaluate and enforce them. This table is intended as a quick-start checklist: an engineer beginning a new design should identify the applicable invariants, run M1, and configure M2 before launching any optimizer.

Table 10: Available invariants: implementation status and module mapping.

AI	**Name**	**Category**	**Type**	**C3 v1**	**Module**
1	Etendue	Geometric	Hard	Implemented	M1, M2
2	Lagrange	Geometric	Hard	Implemented	M1, M2
3	Diffraction limit	Wave-optical	Hard	Implemented	M1, M2
4	SBP	Wave-optical	Soft	Implemented	M1, M2
5	Petzval sum	Wave-optical	Soft	Implemented	M1, M2
6	Unitarity	Network	Hard	Implemented	M1, M2
7	Reciprocity	Network	Soft	Implemented	M1, M2
8	Topological charge	Topological	Hard	Implemented	M1, M2
9	$S = F$ threshold	Quantum	Hard	v2 planned	M2 (via bridge)
10	Decoherence rate	Quantum	Hard	v2 planned	—
11	Entanglement E_F	Quantum	Hard	v2 planned	—
12	Schmidt number K	Quantum	Soft	v2 planned	—
13	Cooperativity C	Quantum	Hard	v2 planned	—
14	Brody β	Quantum/Chaos	Soft	v2 planned	—
15	Fidelity $\mathcal{F}$	Quantum	Hard	v2 planned	—

13.2 Configuration Optimization: Landscape and Implementation Roadmap

The Invariant Gate's configuration discovery benefit — finding *different* designs, not just the same design faster — applies broadly across optical engineering. Table 11 surveys the landscape: for each application domain, it identifies which invariants are likely to be binding, what kind of configuration change the gate can enable, the expected impact level, and a concrete implementation pathway.

The table distinguishes three mechanisms by which invariant constraints enable configuration discovery:

1. **Basin selection:** the gate eliminates infeasible basins, directing the optimizer toward a different (and better) basin of attraction. This is the TC1 mechanism.
2. **Co-optimization:** the gate couples degrees of freedom that standard optimizers treat independently (e.g., phase and thermal power), creating a joint landscape with different optima. This is the TC2 mechanism.
3. **Landscape compression:** tensor decomposition removes redundant dimensions, collapsing the high-dimensional landscape into a lower-dimensional space where global structure is more accessible. This is the TC3 mechanism.

Table 11: Configuration optimization landscape: application domains, binding invariants, configuration change mechanism, impact level (H/M/L), and implementation pathway.

Application domain	**Binding AI**	**Mech.**	**Configuration change enabled**	**Impact**	**Implementation**
Classical imaging					
Double Gauss redesign	AI-1, AI-5	Basin	Different Seidel balance; lower FC at edge	H	C3 TC1 (this work)
Wide-angle fisheye	AI-1, AI-2	Basin	Different element bending to maintain etendue across $> 90^\circ$ FOV	H	Extend TC1 to $> 90^\circ$ FOV
Freeform imaging	AI-4, AI-5	Compress	Tensor-compressed Zernike/XY polynomial coefficients; fewer spurious minima	H	Combine M4 with M1 (AI-4 gate)
Microscope objective	AI-1, AI-3	Basin	High-NA immersion designs that unconstrained optimizer avoids due to glass-boundary infeasibility	M	M1 audit at NA $>$ 1.0
Datacenter / telecom photonics					
800G MZI mesh	AI-6, thermal	Co-opt	Joint phase+power allocation; thermally stable configurations	H	C3 TC2 (this work)
Microring filter bank	AI-6, thermal	Co-opt	Resonance-aware heater placement; avoid thermal resonance locking	H	Extend TC2 to ring resonators

Application domain	**Binding AI**	**Mech.**	**Configuration change enabled**	**Impact**	**Implementation**
Arrayed waveguide grating	AI-1, AI-6	Basin	Path-length matching under correlated fab disorder; different arm ordering	M	M1 (AI-1 for grating etendue) + C2 tolerance feedback
Co-packaged optics	AI-6, thermal	Co-opt	Die-level heater layout co-optimized with channel assignment	H	M5 extended to multi-device thermal model (C6)
Quantum photonics					
Metasurface for QI	AI-3, AI-4	Compress	Tucker-compressed high-parameter optimization with $S = F$ verification	H	C3 TC3 (this work)
Boson sampling mesh	AI-6, AI-15	Basin	Unitary mesh configurations with Haar-random fidelity > 0.99; avoids lossy basins	H	M1 (AI-6) + B5 fidelity gate
QKD transmitter	AI-8, AI-9	Basin	OAM mode allocation respecting topological charge; avoids decoherence-limited configs	M	M1 (AI-8) + C4 S=F threshold
Entangled-photon source	AI-11, AI-13	Basin	Cavity design respecting cooperativity gate; different mode-matching topology	M	M1 (AI-13 when v2) + B4 cooperativity
Illumination					
DMD projector	AI-1	Basin	Source-DMD etendue matching; different illumination architecture (Class I vs II)	M	M1 etendue audit (B3 Case 1)
Structured illumination	AI-1, AI-8	Basin	OAM-based illumination patterns respecting charge conservation	M	M1 (AI-1, AI-8)
High-power LED thermal	Thermal	Co-opt	Junction temperature co-optimized with luminous flux distribution	H	M5 thermal engine + A3 Mode-Path Matrix
Sensing					
MZI phase sensor	AI-9	Basin	Quantum-fidelity-constrained MZI design; different arm-length ratio for squeezed-state operation	H	M1 ($S = F$ threshold) + C4 M1

Application domain	Binding AI	Mech.	Configuration change enabled	Impact	Implementation
Fiber gyroscope	AI-2, AI-7	Basin	Reciprocity-constrained Sagnac design; avoids non-reciprocal bias configurations	M	M1 (AI-7 reciprocity audit)
Wavefront sensor	AI-4	Basin	SBP-constrained lenslet design; different sub-aperture tiling	M	M1 (AI-4 SBP audit)

13.3 Configuration Optimization in the Three Test Cases: Detailed Analysis

The three test cases in this document provide concrete demonstrations of the three configuration discovery mechanisms. This subsection consolidates the evidence from each.

13.3.1 TC1: Basin Selection in the Double Gauss

What unconstrained optimization does: The unconstrained optimizer in TC1 starts from a perturbed nominal Double Gauss and converges to a local minimum in the 12-dimensional design space. This minimum has a characteristic Seidel signature: moderate spherical aberration (SA) and coma (CO), but relatively high field curvature (FC) and astigmatism (AS) at the field edge. The reason is that the unconstrained optimizer's path to the global minimum passes through an etendue-violating region near 70% field — a "desert" in the search landscape where no physical design exists. The optimizer cannot cross this desert, so it settles for the nearest local minimum on its side.

What the Invariant Gate changes: By enforcing AI-1 (etendue barrier) and AI-5 (Petzval penalty), the gate removes the infeasible desert and the local minima that exist within or adjacent to it. The constrained optimizer's landscape is topologically different: the basins of attraction are rearranged. The optimizer converges to a configuration with lower Petzval sum, lower field curvature, and a different element-bending balance — a design form that the unconstrained optimizer could not reach.

Evidence: Figure 9(c) shows the Seidel coefficient comparison. The constrained solution has a distinctly different FC/AS ratio, confirming that the two optimizers converged to different lens forms, not different parameterizations of the same form.

Generalization: This basin-selection mechanism applies to any multi-element lens system where the etendue margin varies across the field. Wide-angle systems (FOV $> 60°$), high-NA microscope objectives, and anamorphic cinema lenses are all candidates. In each case, the binding invariant (AI-1 or AI-2) creates infeasible deserts that trap unconstrained optimizers.

Table 12: TC1 configuration optimization detail.

	Unconstrained	**Invariant-constrained**
Convergence target	Nearest local minimum (same side of etendue desert)	Basin accessible only after desert is removed
Seidel balance	High FC and AS at edge; moderate SA	Lower FC, redistributed AS; different element bending
Edge Strehl	Limited by field curvature floor	Improved (Petzval penalty reduces FC)
Design form	Standard Double Gauss with nominal bending	Modified bending with Petzval-balanced glass selection
Mechanism	—	**Basin selection** via AI-1 barrier + AI-5 penalty

13.3.2 TC2: Co-optimization in the MZI Mesh

What unconstrained optimization does: The thermal-unaware optimizer in TC2 treats the 16 MZI phase shifts as the only design variables. It finds a set of phases that minimizes crosstalk at the *nominal* temperature. However, when thermal crosstalk from adjacent heaters is accounted for, the actual phases differ from the intended values by 5–15%, and the realized isolation degrades.

What the Invariant Gate changes: The thermal-aware optimizer (M5) treats the thermal crosstalk matrix K_{ij} as a first-class constraint. The merit function becomes a *joint* function of both the intended phases and the heater power allocation. This joint landscape has a different structure: some phase configurations that are optimal at nominal temperature become suboptimal when thermal stability is included, because they require large heater powers that generate excessive crosstalk. The optimizer finds a *different* phase configuration — one that trades a small amount of nominal isolation for much better thermal stability.

Evidence: Figure 16(a) shows per-channel isolation for both approaches. The thermal-aware solution has more uniform isolation across channels, indicating a fundamentally different phase allocation strategy.

Generalization: The co-optimization mechanism applies to any PIC where thermo-optic tuning is used: microring filter banks, arrayed waveguide gratings, Mach–Zehnder delay lines, and co-packaged optics modules. In each case, the "thermal budget" acts as an emergent invariant that couples the phase design space to the power allocation space.

Table 13: TC2 configuration optimization detail.

	Thermal-unaware	**Thermal-aware**
Design variables	Phase shifts only (16 DOFs)	Phase shifts + heater powers (16 + 8 DOFs)
Landscape	Phase-only surface; thermal effects are post-hoc	Joint phase-power surface; thermal crosstalk built in
Optimal config	Best nominal isolation; poor thermal stability	Slightly lower nominal isolation; much better stability
Channel uniformity	Non-uniform (edge channels worse)	More uniform (power balancing equalizes channels)
Mechanism	—	**Co-optimization** via M5 thermal constraint + AI-6 gate

13.3.3 TC3: Landscape Compression in the Metasurface

What unconstrained optimization does: The full-evaluation optimizer in TC3 operates on a 500-dimensional landscape (one phase per meta-atom, demonstration scale). At production scale ($N = 10^4$), the $O(N^2)$ merit function evaluation makes each iteration take minutes. But the computational challenge is not just speed: the high-dimensional landscape contains many spurious local minima arising from the pairwise coupling between distant meta-atoms whose interaction is physically negligible.

What the Invariant Gate changes: Tucker decomposition (M4) projects the $N \times N$ interaction matrix onto a rank-R subspace ($R \ll N$). This is not merely a speedup trick — it is a *dimensionality reduction* of the effective landscape. Interactions between distant atoms (which contribute to spurious minima) are captured in the discarded singular values. The compressed landscape has fewer local minima and its basins are broader, making gradient-based optimization more reliable. The AI-4 (SBP) gate ensures that the compression does not discard physically meaningful degrees of freedom: if SBP $< N$, some meta-atoms are redundant and compression is not only safe but physically motivated.

Evidence: Figure 21 shows that compressed and full optimization converge to similar final merit function values, but the compressed optimizer's convergence is smoother (fewer stalls at local minima). The scaling study (Fig. 24) confirms that the speedup grows with N, projecting $> 100\times$ at production scale.

Generalization: The landscape compression mechanism applies to any design problem with a large number of parameters and a pairwise interaction structure: diffractive optical elements, photonic crystal slabs, phased arrays, and multi-layer thin-film stacks. In each case, the interaction matrix has a rapidly decaying singular value spectrum (because physical interactions are local), making low-rank compression both accurate and physically meaningful.

Table 14: TC3 configuration optimization detail.

	Full evaluation	**Tensor-compressed**
Landscape dimension	N (full parameter space)	$R \ll N$ (projected subspace)
Spurious minima	Many (from weak long-range couplings)	Fewer (weak couplings discarded by truncation)
Convergence	Stalls at local minima; restarts needed	Smoother; broader basins in compressed space
Speedup	$O(N^2)$ per eval	$O(NR + R^2)$; $> 100\times$ at $N = 10^4$
Physical justification	—	AI-4 (SBP): if SBP $< N$, some DOFs are redundant; compression is physically motivated
Mechanism	—	**Landscape compression** via M4 Tucker + AI-4 gate

13.4 Closing Remarks

The Invariant Gate architecture introduced in C3 provides a systematic answer to a question that optical engineers have long addressed *ad hoc*: "Which constraints should I impose before optimization?" The invariant catalog (Table 5) and the configuration landscape survey (Table 11) provide a practitioner-ready framework for answering this question across classical imaging, datacenter photonics, quantum devices, illumination, and sensing.

The deeper insight is that "configuration" should be understood not as a set of geometric parameters but as a *flow topology* — the pattern of etendue allocation, information distribution,

and energy balance through the system (Sec. 1.7). The engineer's practical goal is the configuration whose flow topology is most uniformly distributed and therefore most tolerant of production variability. The Invariant Gate finds these configurations by eliminating flow topologies that are intrinsically fragile — those with concentrated etendue, power hotspots, or information bottlenecks.

The three configuration discovery mechanisms — basin selection (TC1), co-optimization (TC2), and landscape compression (TC3) — are structural consequences of how invariant constraints reshape search landscapes. Any design problem with a binding physical invariant is a candidate for configuration discovery via the Invariant Gate.

C3 v2 will extend the invariant catalog to include quantum invariants (AI-9 through AI-15), add Code V/Zemax API wrappers for production integration, and implement ML surrogate models for merit function acceleration. The configuration discovery mechanisms demonstrated here will serve as the foundation for the D-series worked examples (D3 Imaging, D4 Illumination, D5 Sensing, D6 Communication), where each application domain inherits the Invariant Gate as its pre-optimization checklist.

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